

BFS Traversal

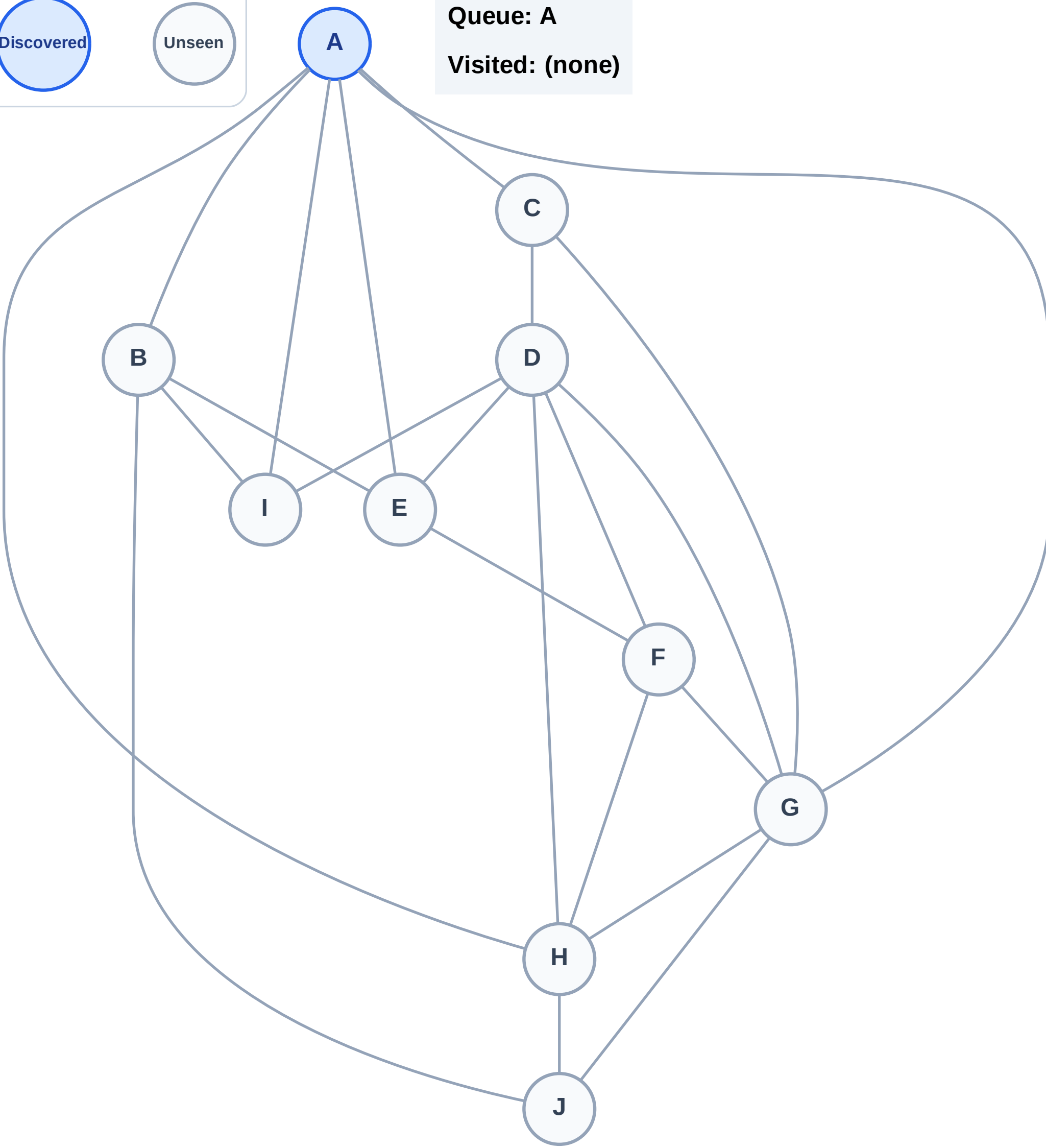
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

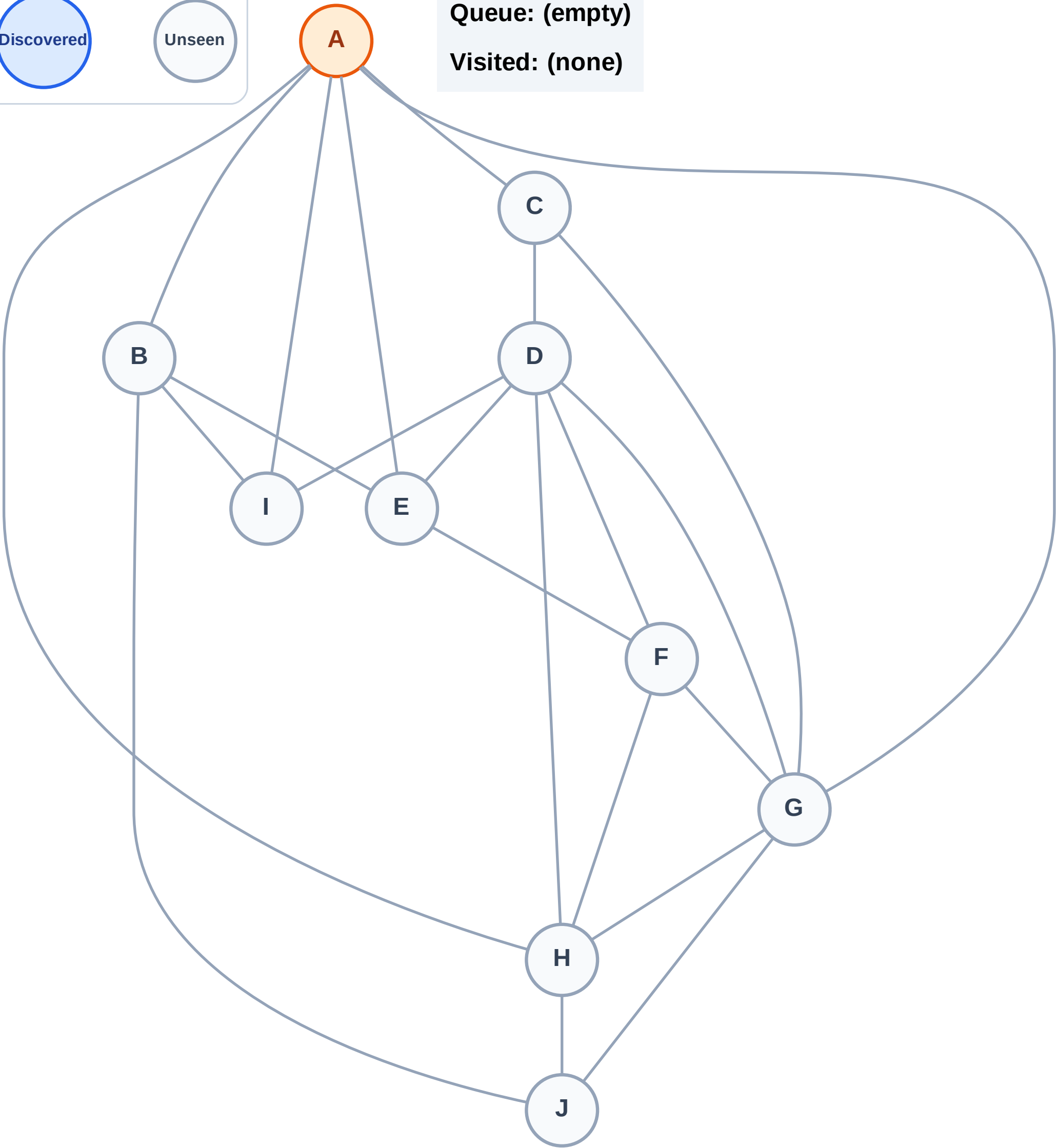
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

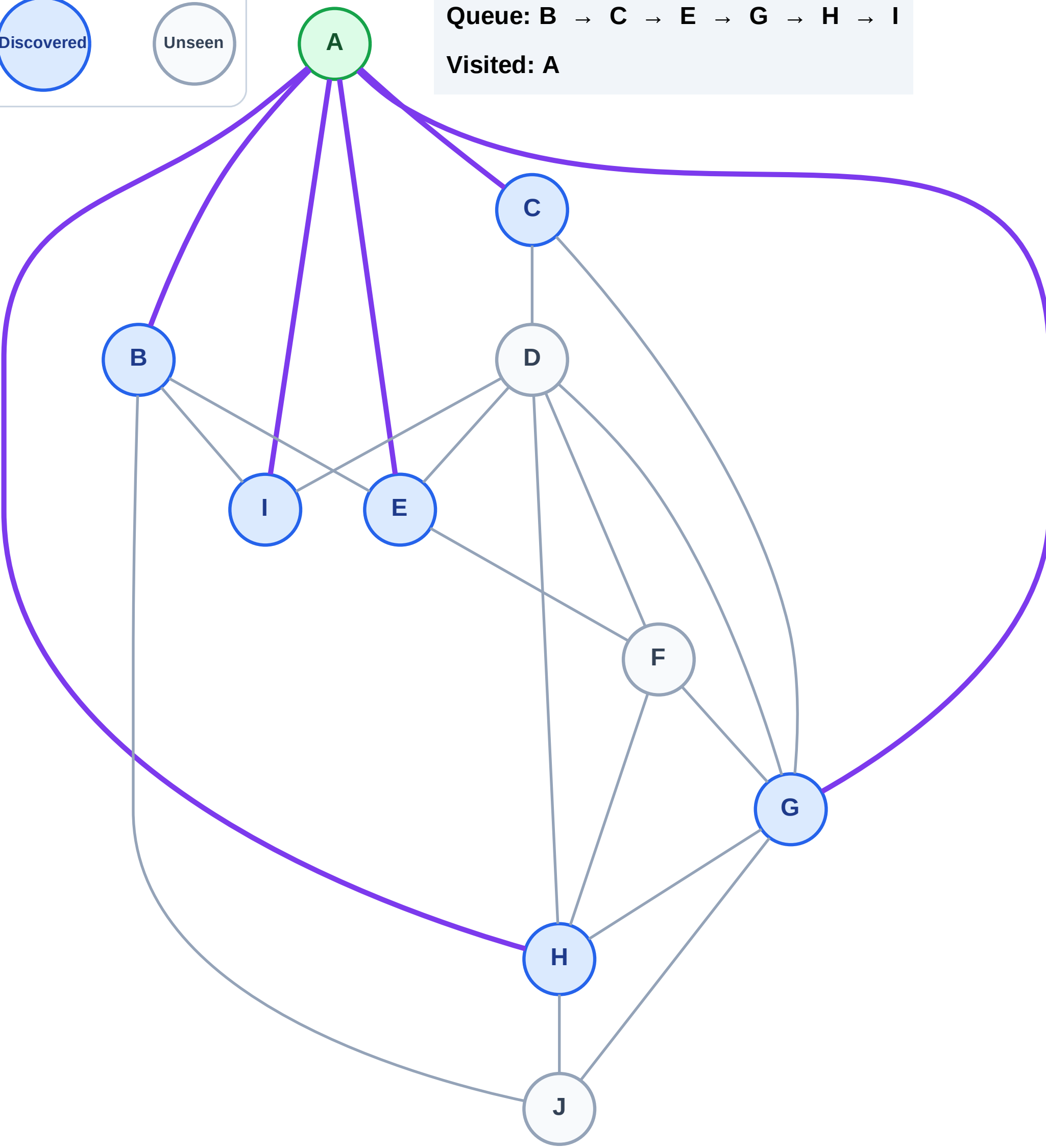
Step 3: Discover B, C, E, G, H, I from A

Legend



Queue: B → C → E → G → H → I

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

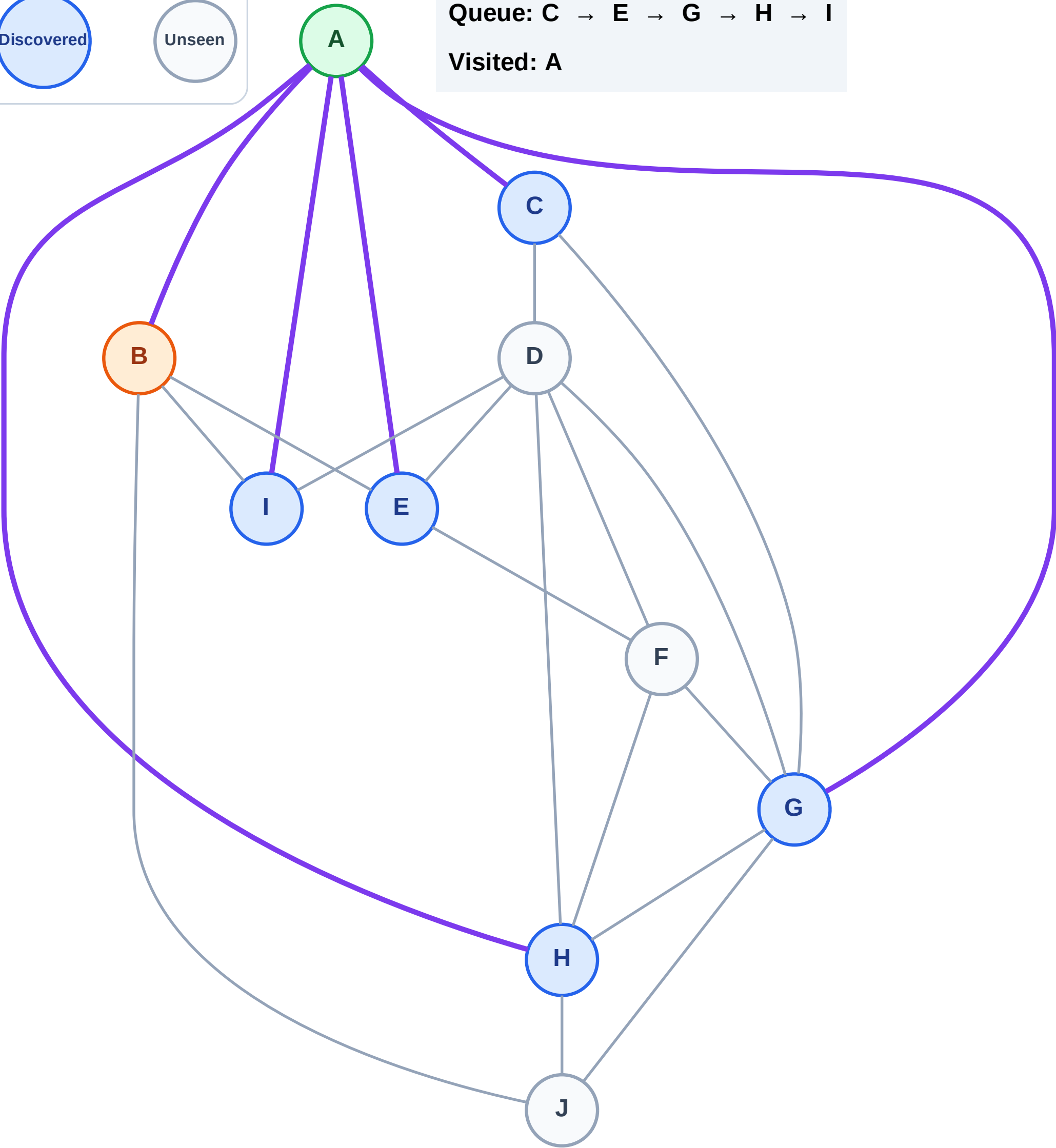
Processing

Discovered

Unseen

Queue: C → E → G → H → I

Visited: A



BFS Traversal

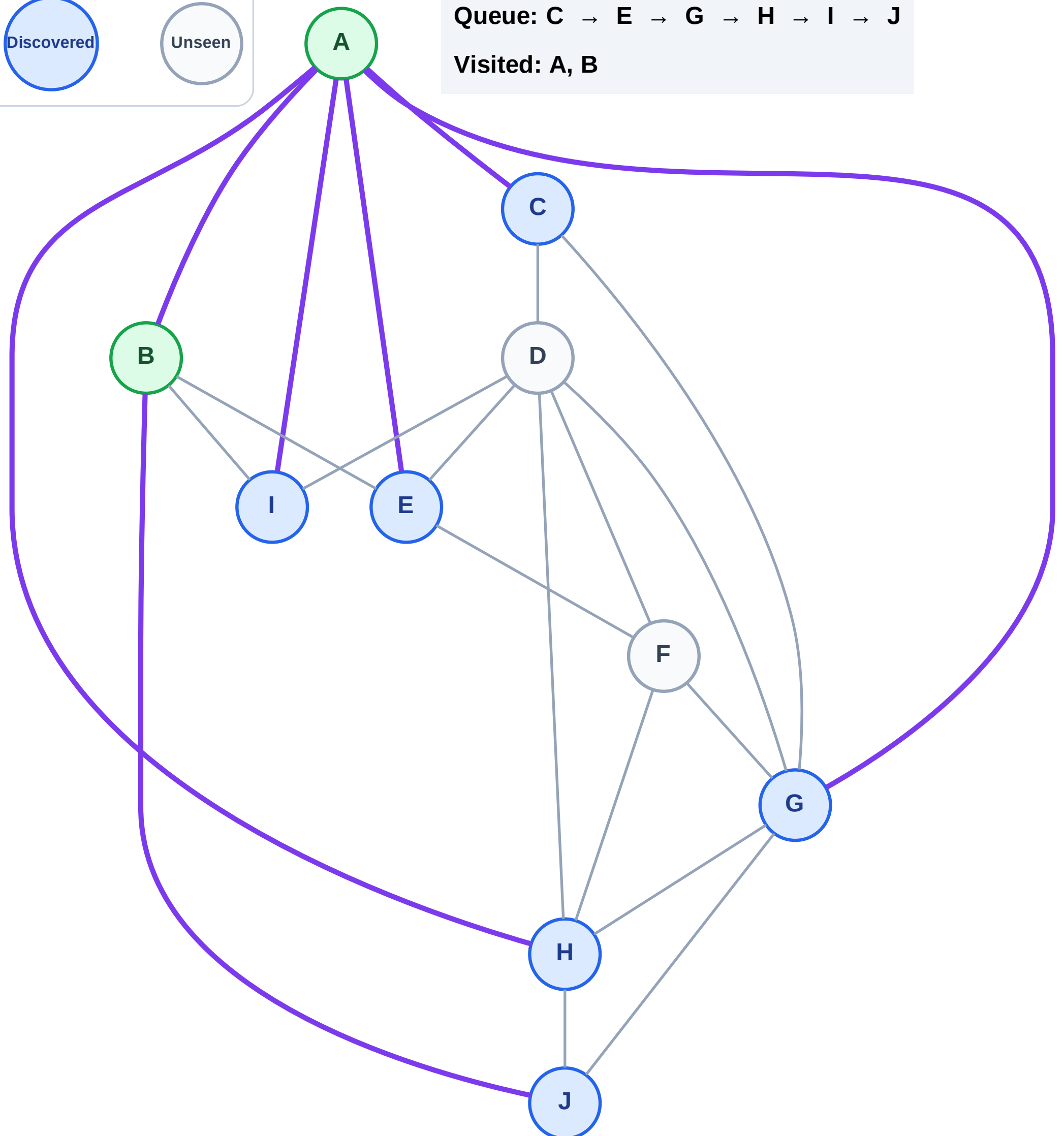
Step 5: Discover J from B

Legend



Queue: C → E → G → H → I → J

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

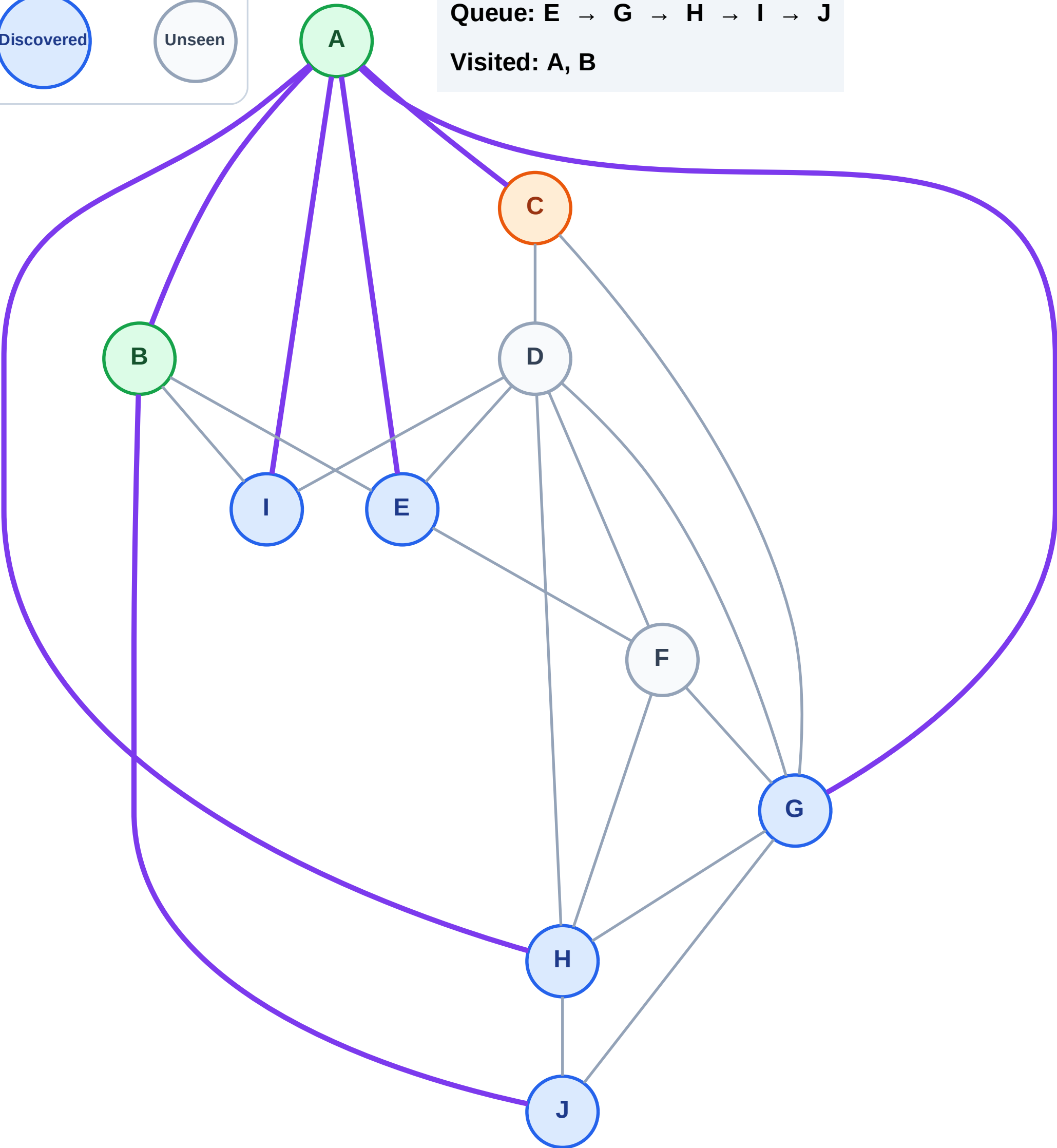
Processing

Discovered

Unseen

Queue: E → G → H → I → J

Visited: A, B



BFS Traversal

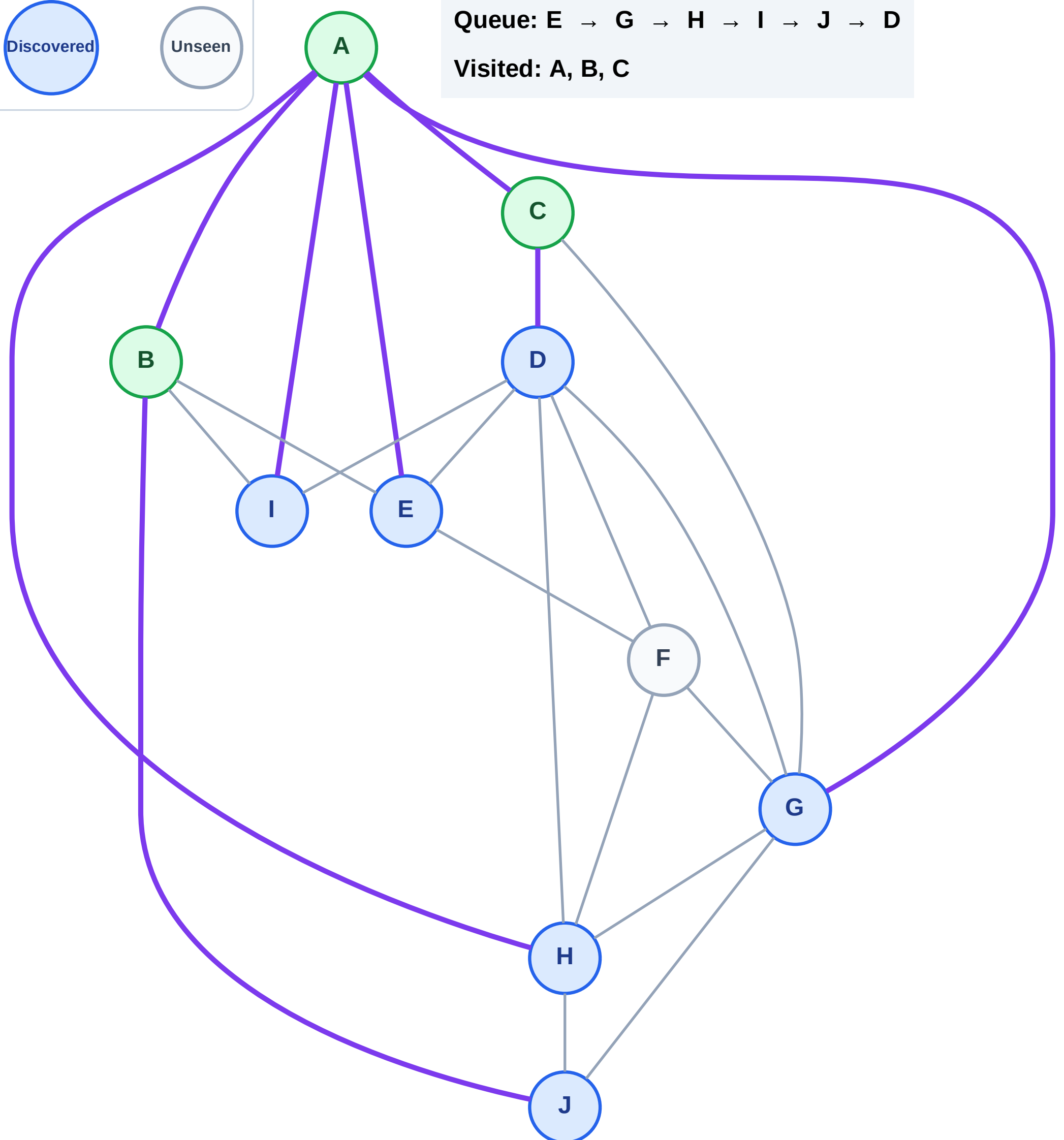
Step 7: Discover D from C

Legend



Queue: E → G → H → I → J → D

Visited: A, B, C



BFS Traversal

Step 8: Process node E

Legend

Complete

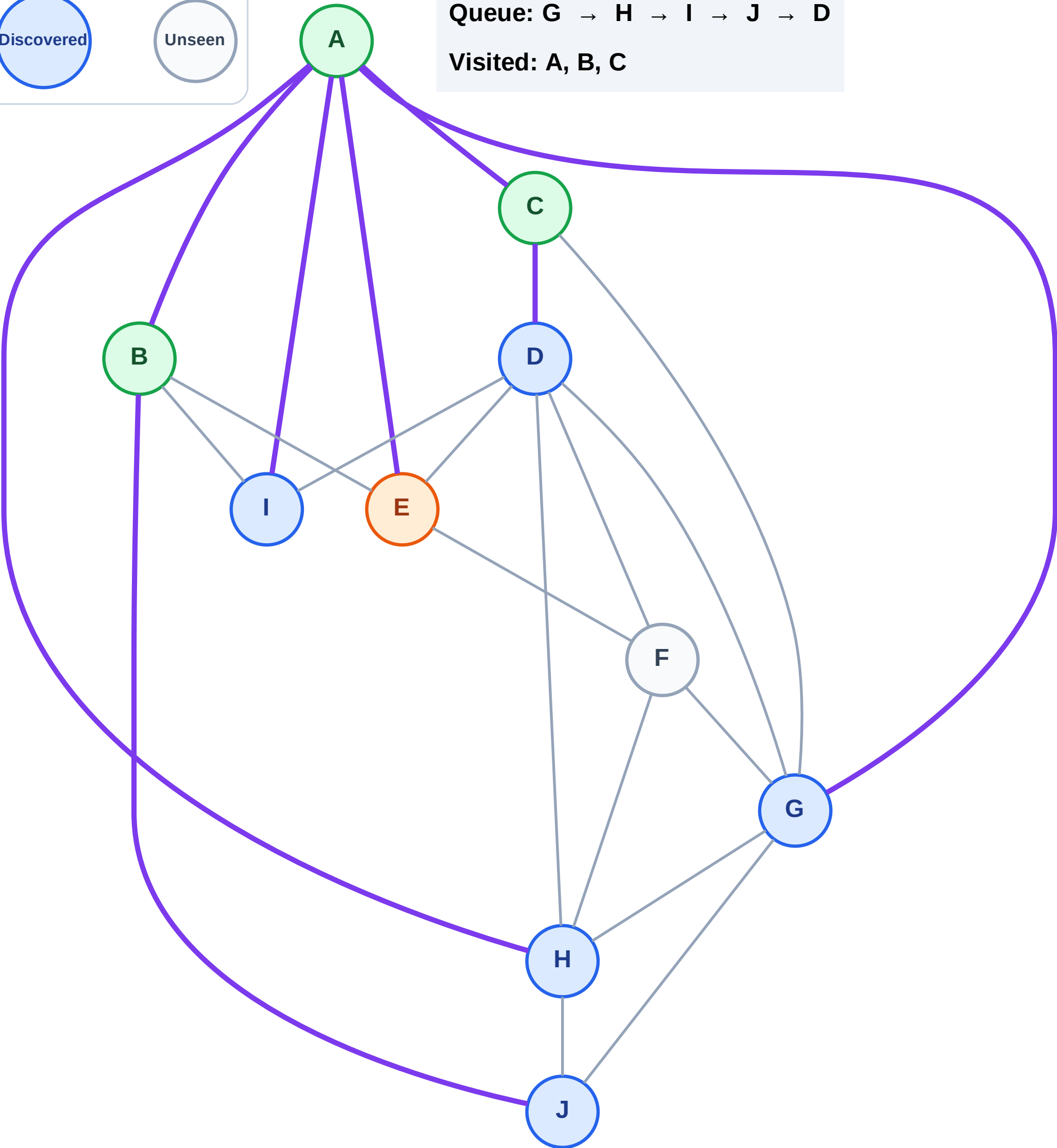
Processing

Discovered

Unseen

Queue: G → H → I → J → D

Visited: A, B, C



BFS Traversal

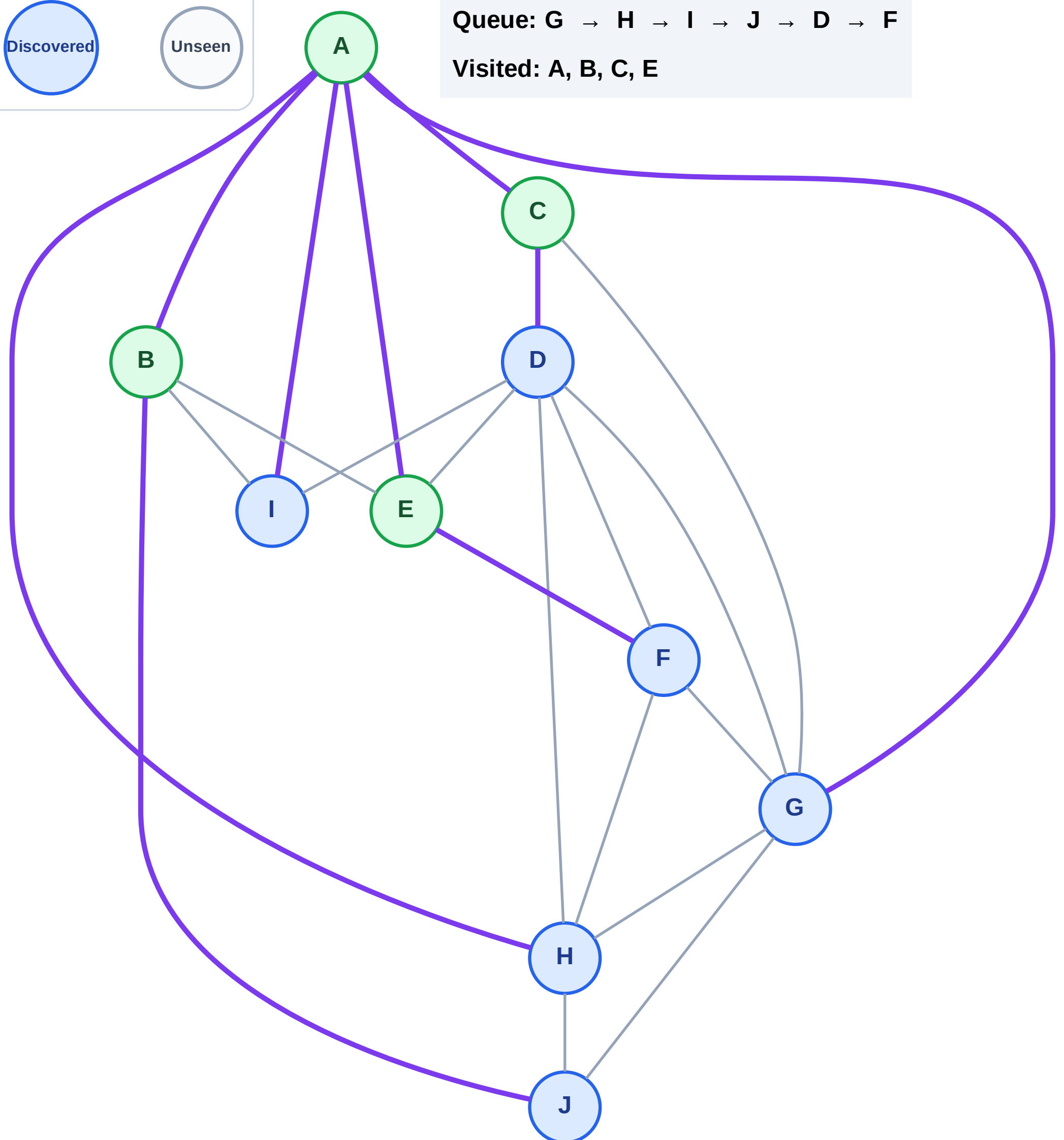
Step 9: Discover F from E

Legend



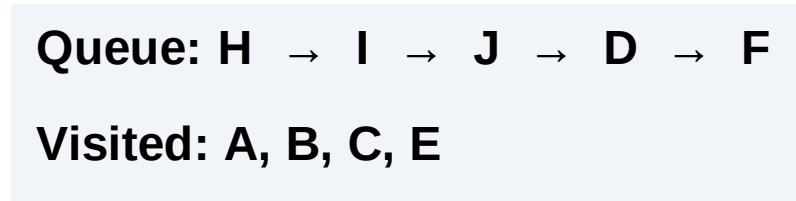
Queue: G → H → I → J → D → F

Visited: A, B, C, E



Step 10: Process node G

Step 10: Process node G



BFS Traversal

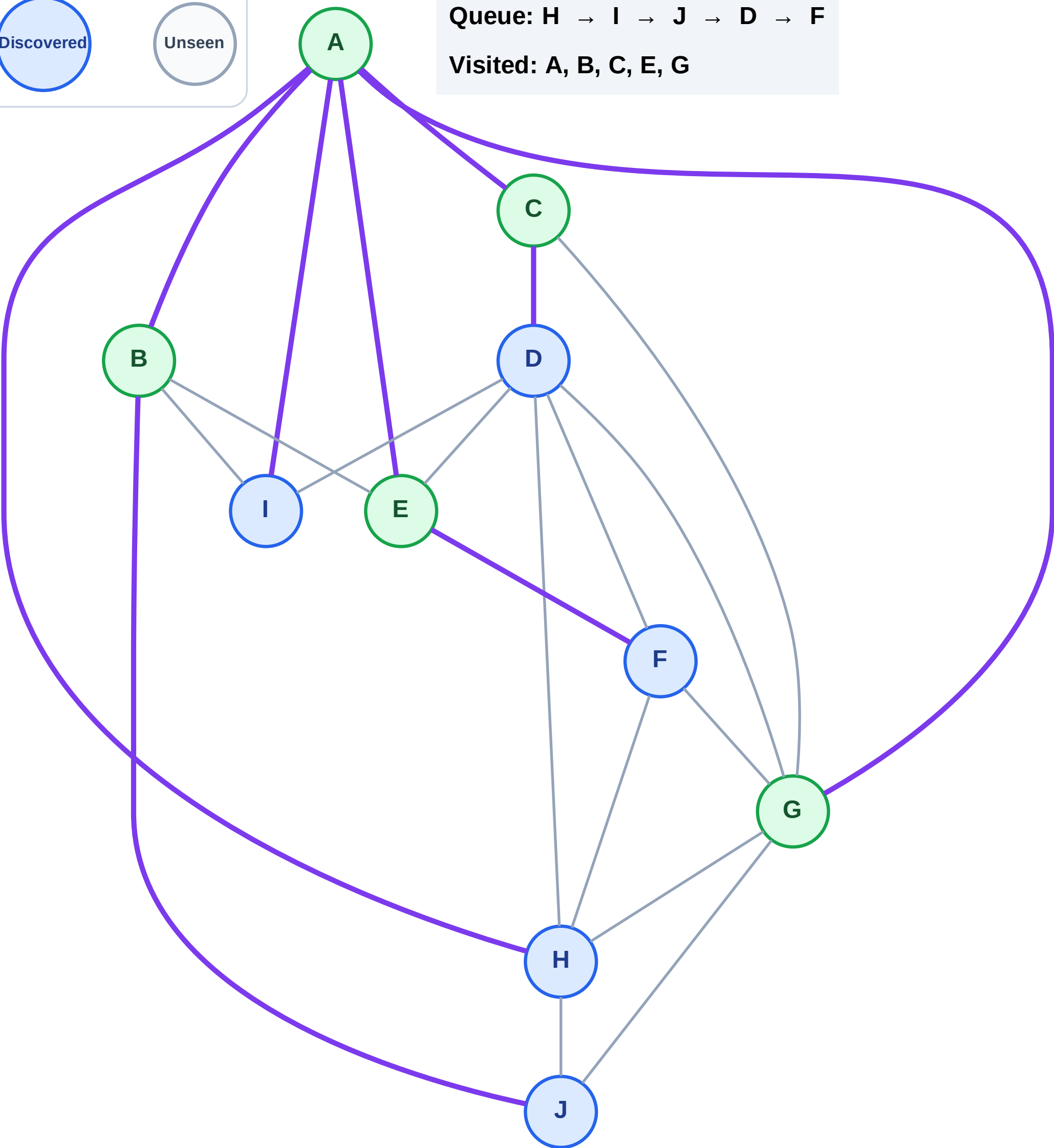
Step 11: No new neighbors from G

Legend



Queue: H → I → J → D → F

Visited: A, B, C, E, G



BFS Traversal

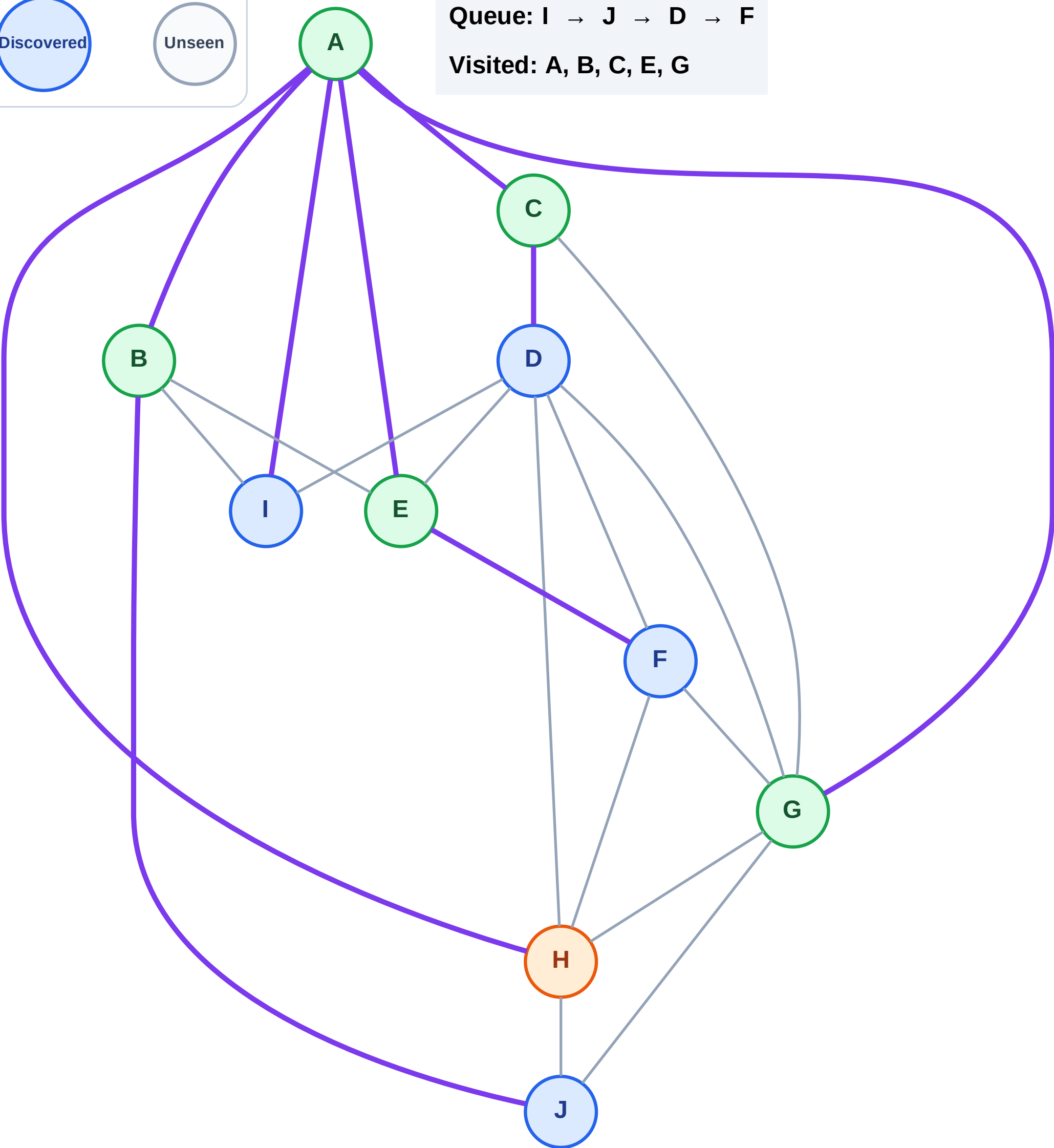
Step 12: Process node H

Legend



Queue: I → J → D → F

Visited: A, B, C, E, G



BFS Traversal

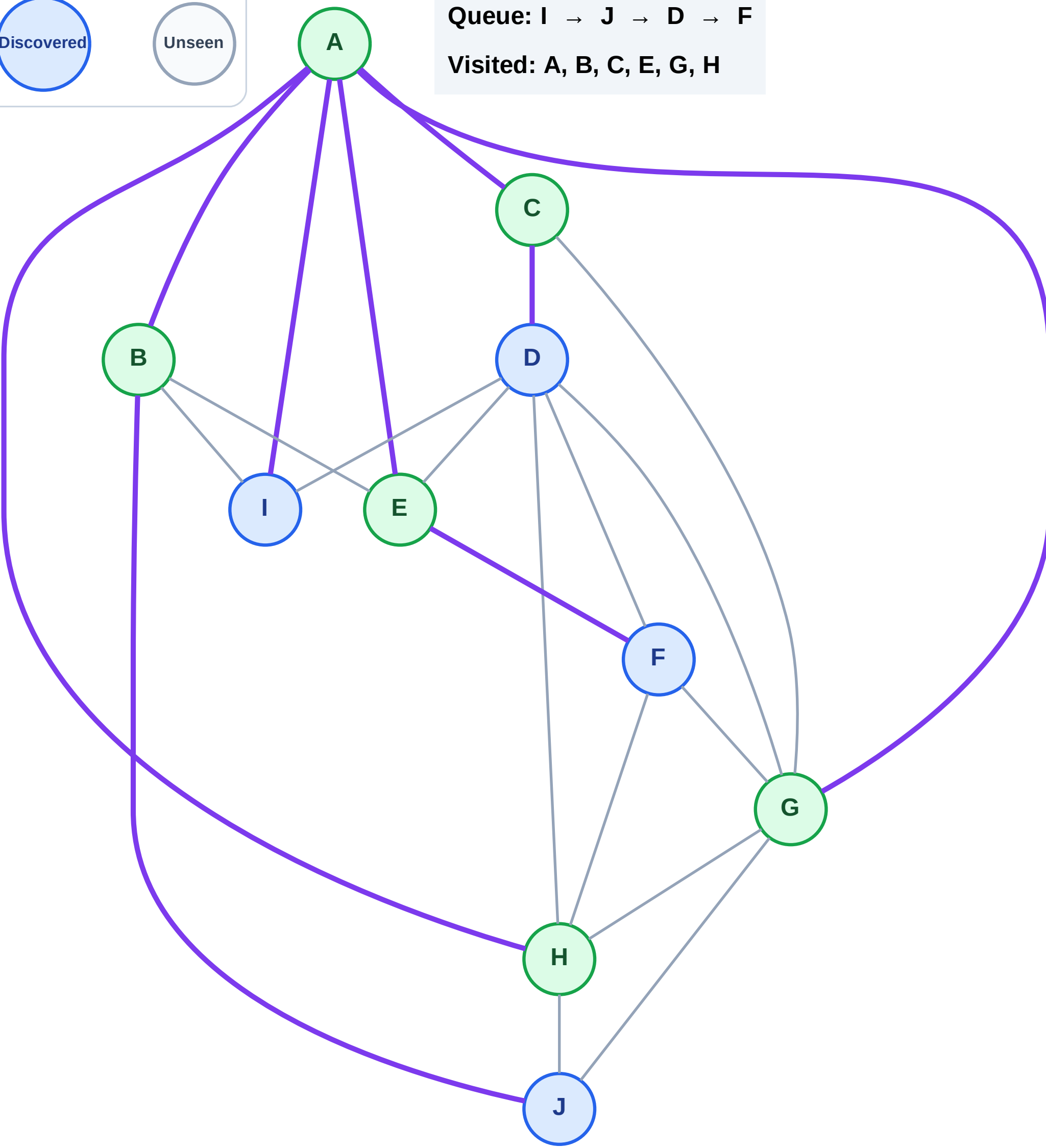
Step 13: No new neighbors from H

Legend



Queue: I → J → D → F

Visited: A, B, C, E, G, H



BFS Traversal

Step 14: Process node I

Legend

Complete

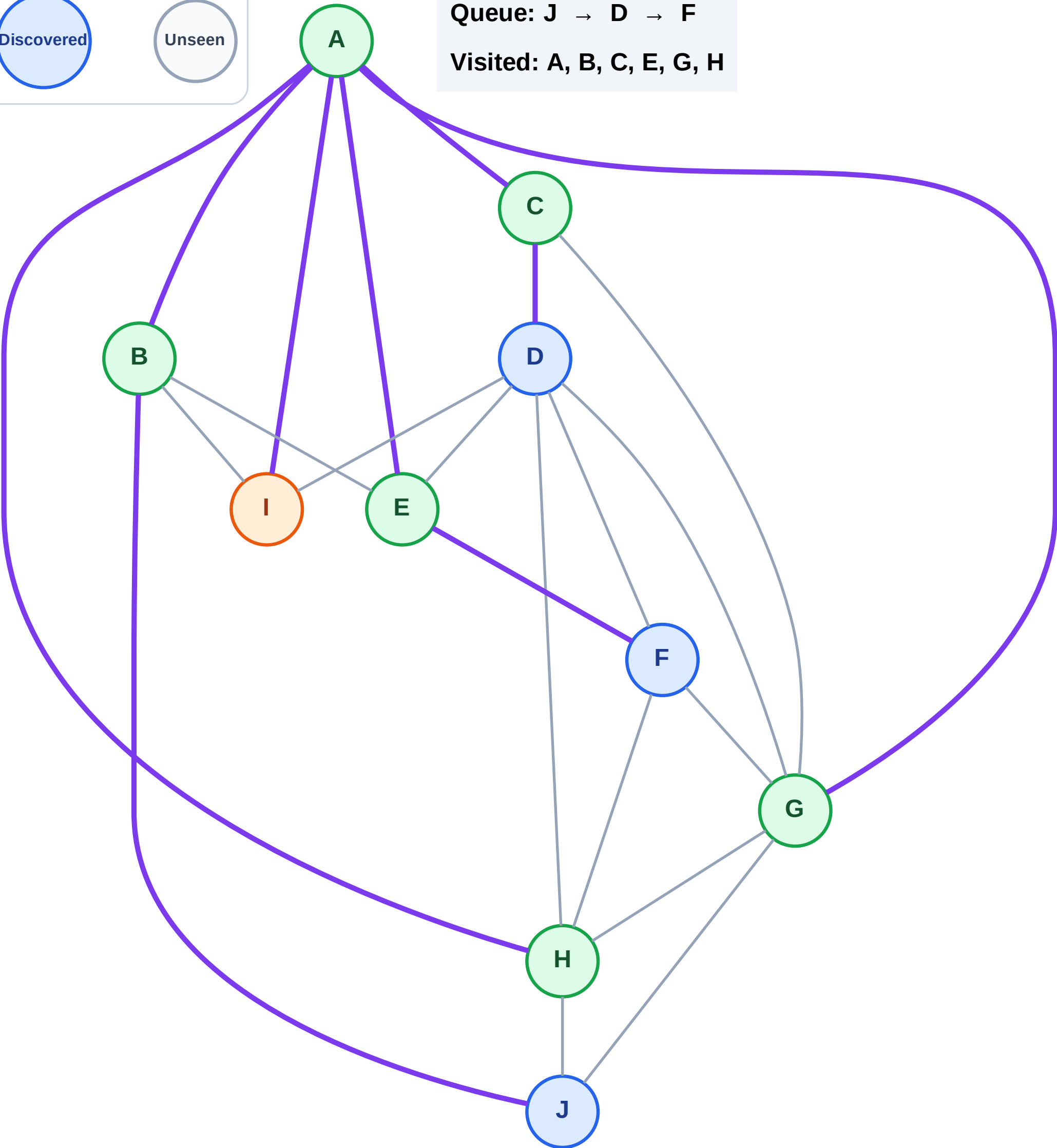
Processing

Discovered

Unseen

Queue: J → D → F

Visited: A, B, C, E, G, H



BFS Traversal

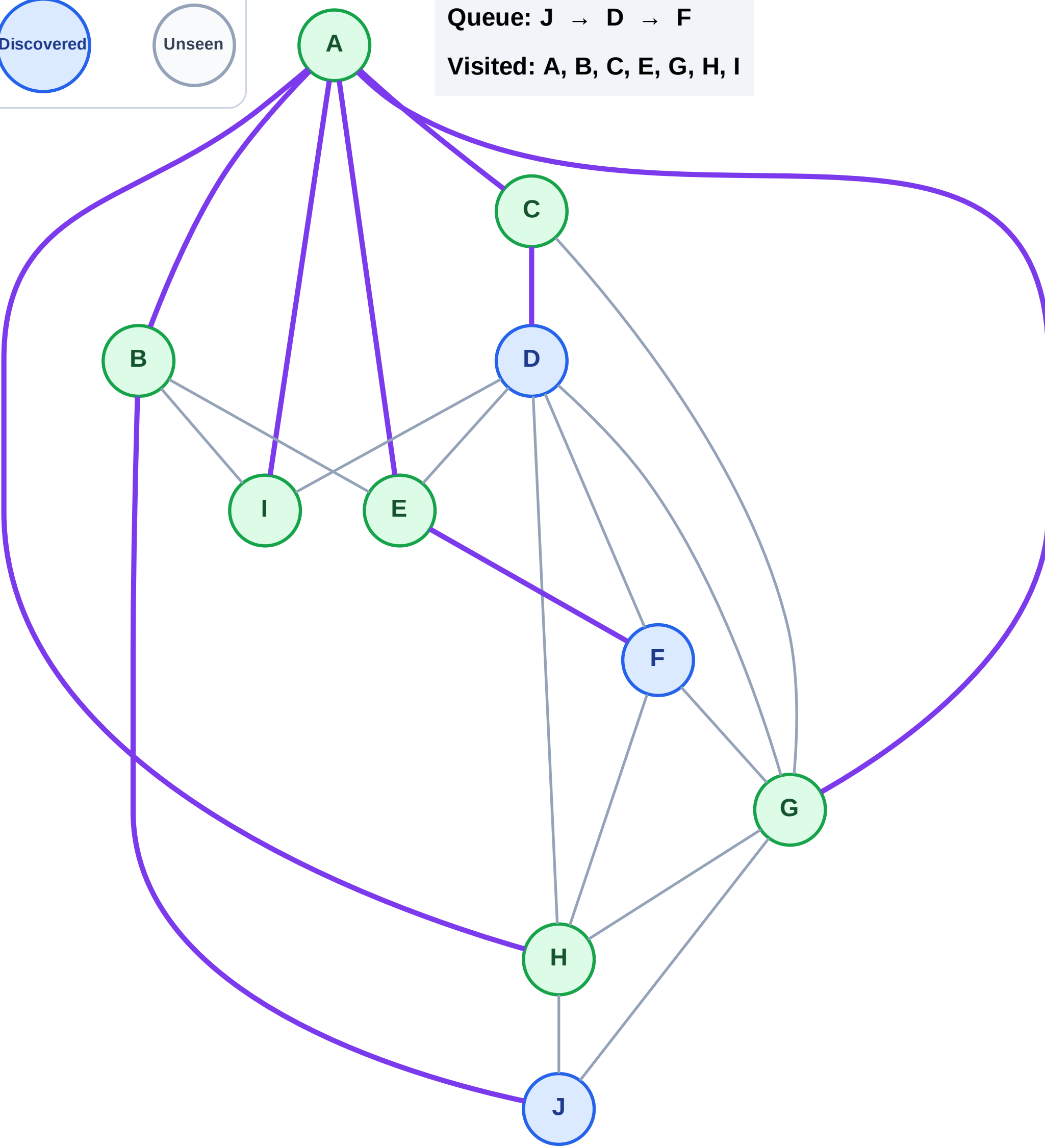
Step 15: No new neighbors from I

Legend



Queue: J → D → F

Visited: A, B, C, E, G, H, I



BFS Traversal

Step 16: Process node J

Legend

Complete

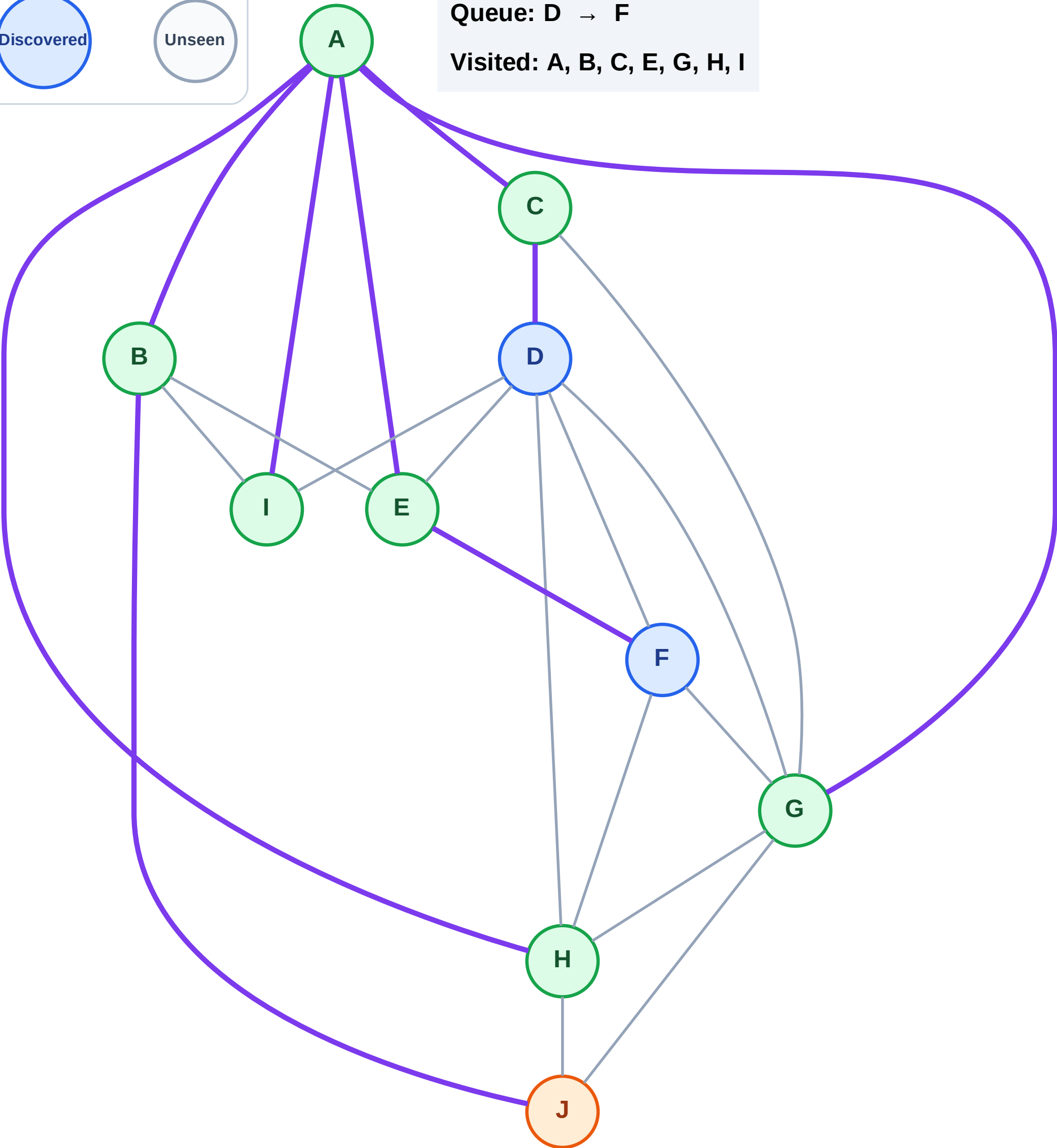
Processing

Discovered

Unseen

Queue: D → F

Visited: A, B, C, E, G, H, I



BFS Traversal

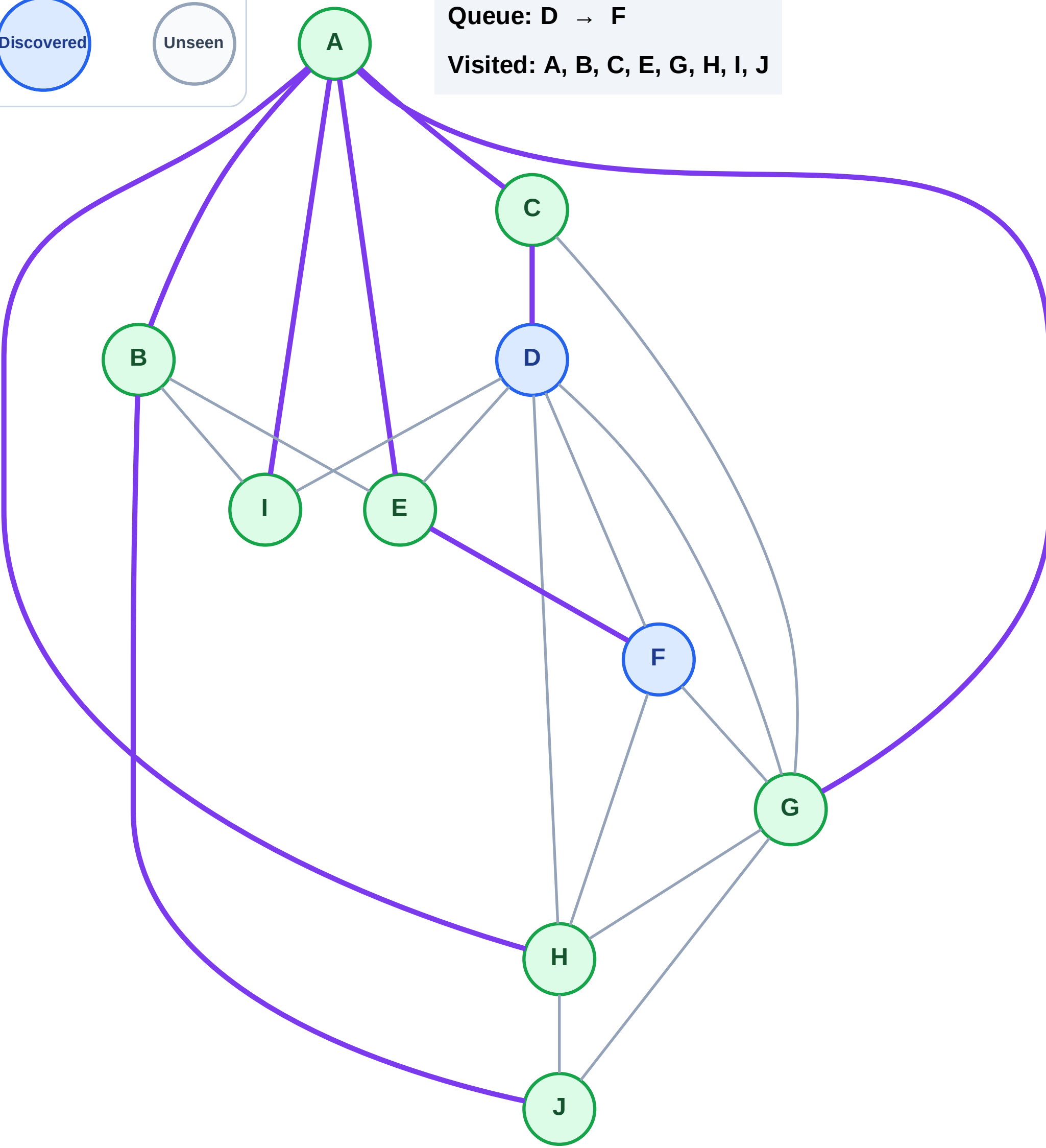
Step 17: No new neighbors from J

Legend



Queue: D → F

Visited: A, B, C, E, G, H, I, J



BFS Traversal

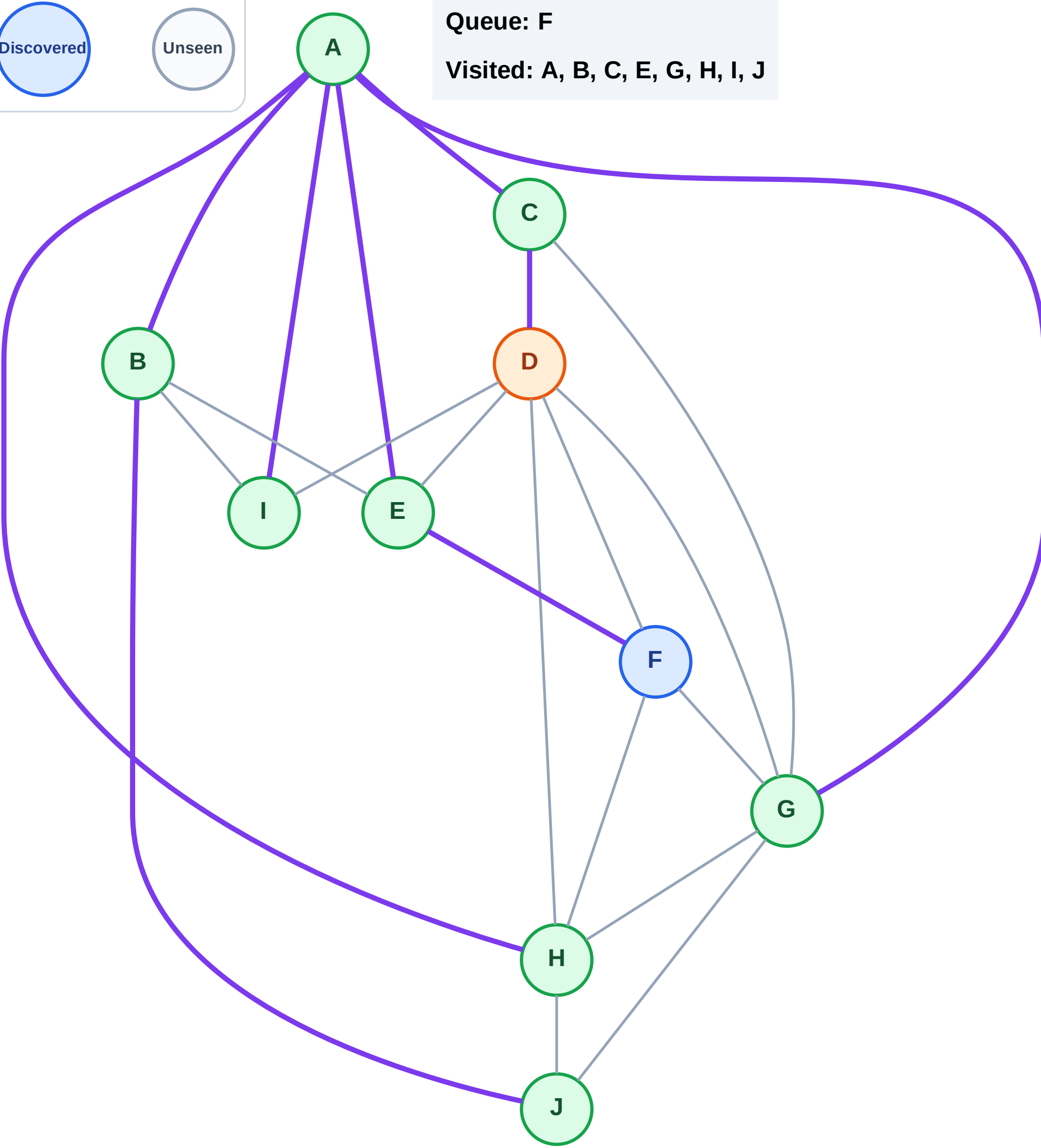
Step 18: Process node D

Legend



Queue: F

Visited: A, B, C, E, G, H, I, J



BFS Traversal

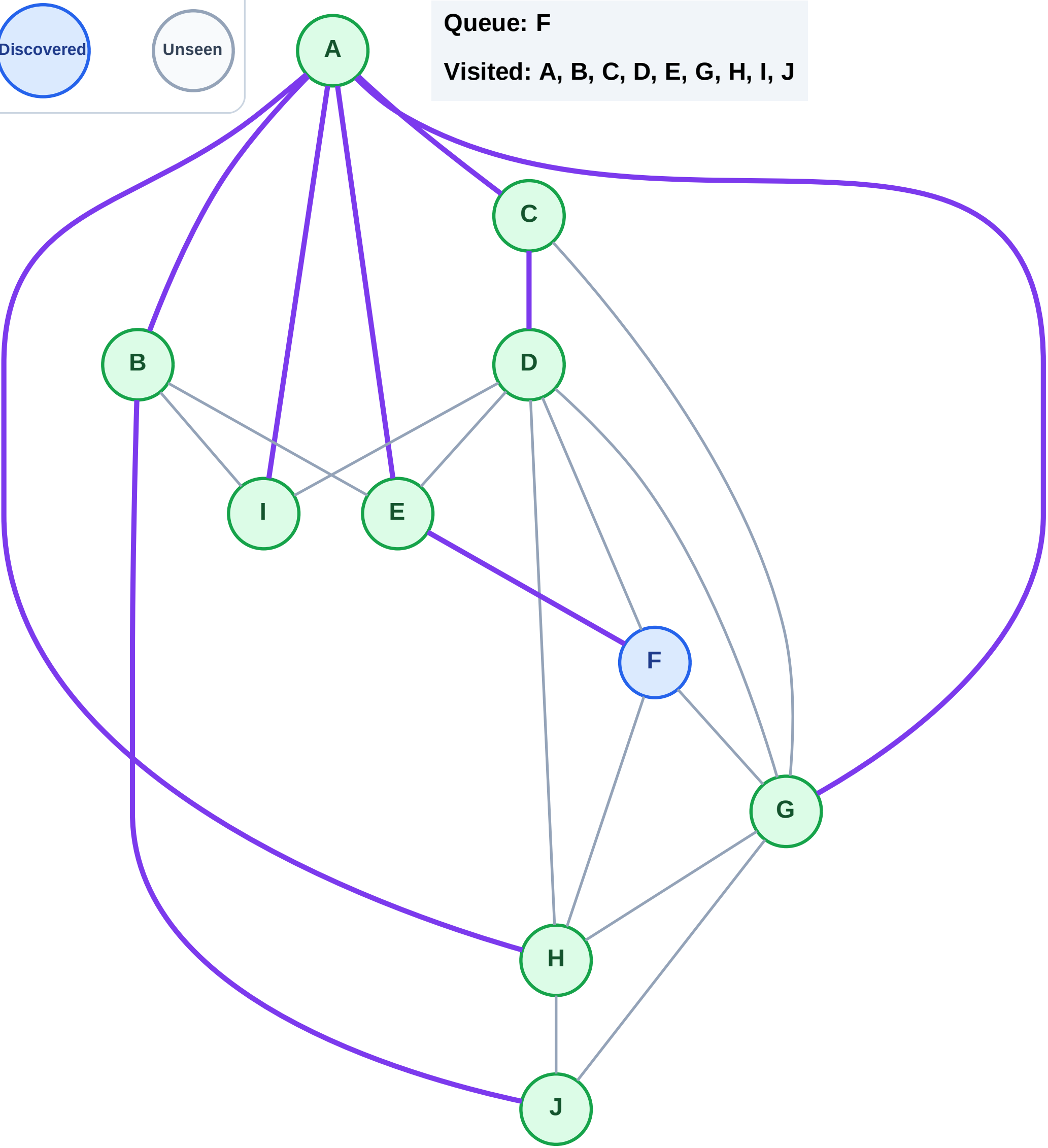
Step 19: No new neighbors from D

Legend



Queue: F

Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

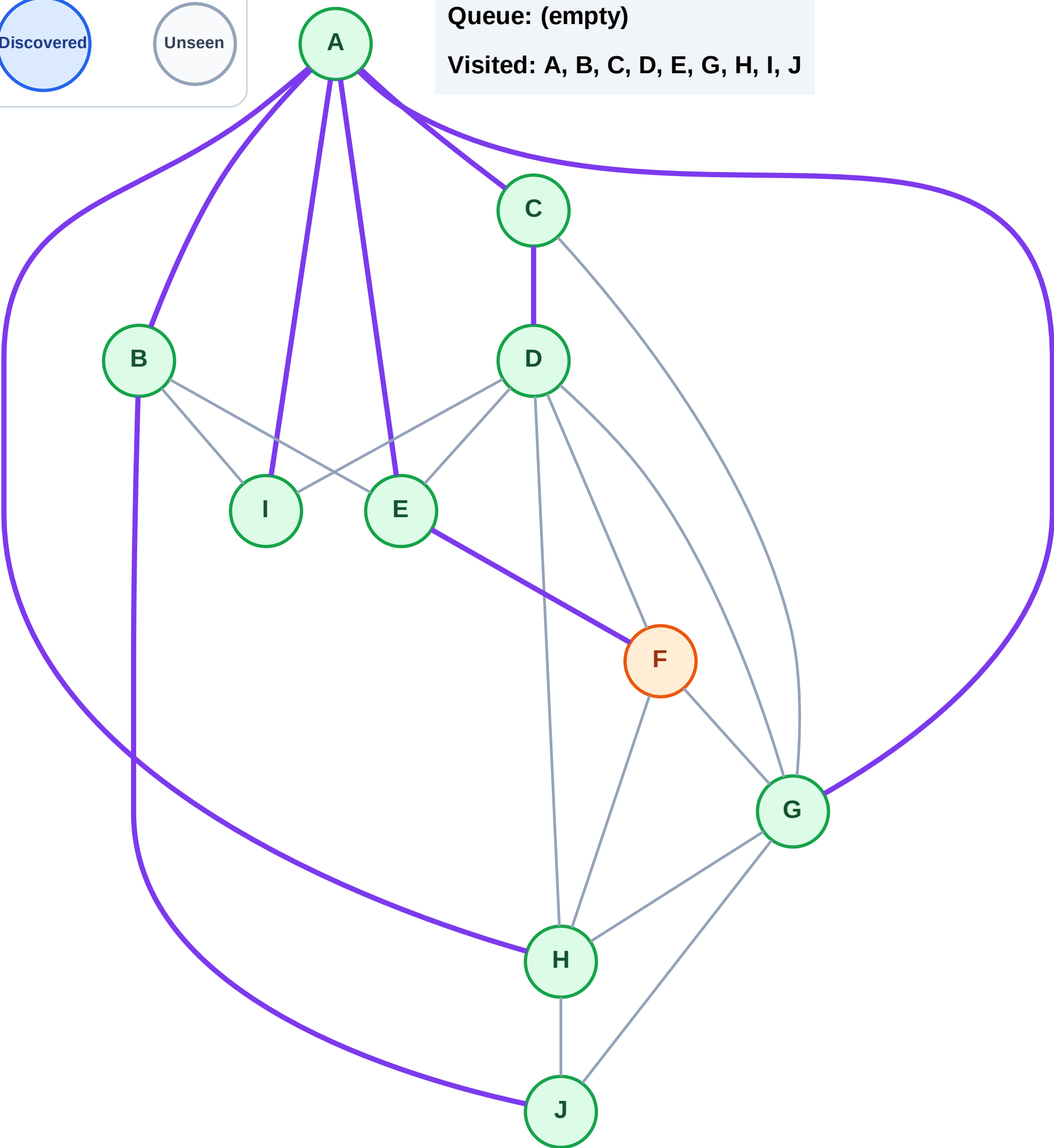
Step 20: Process node F

Legend



Queue: (empty)

Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

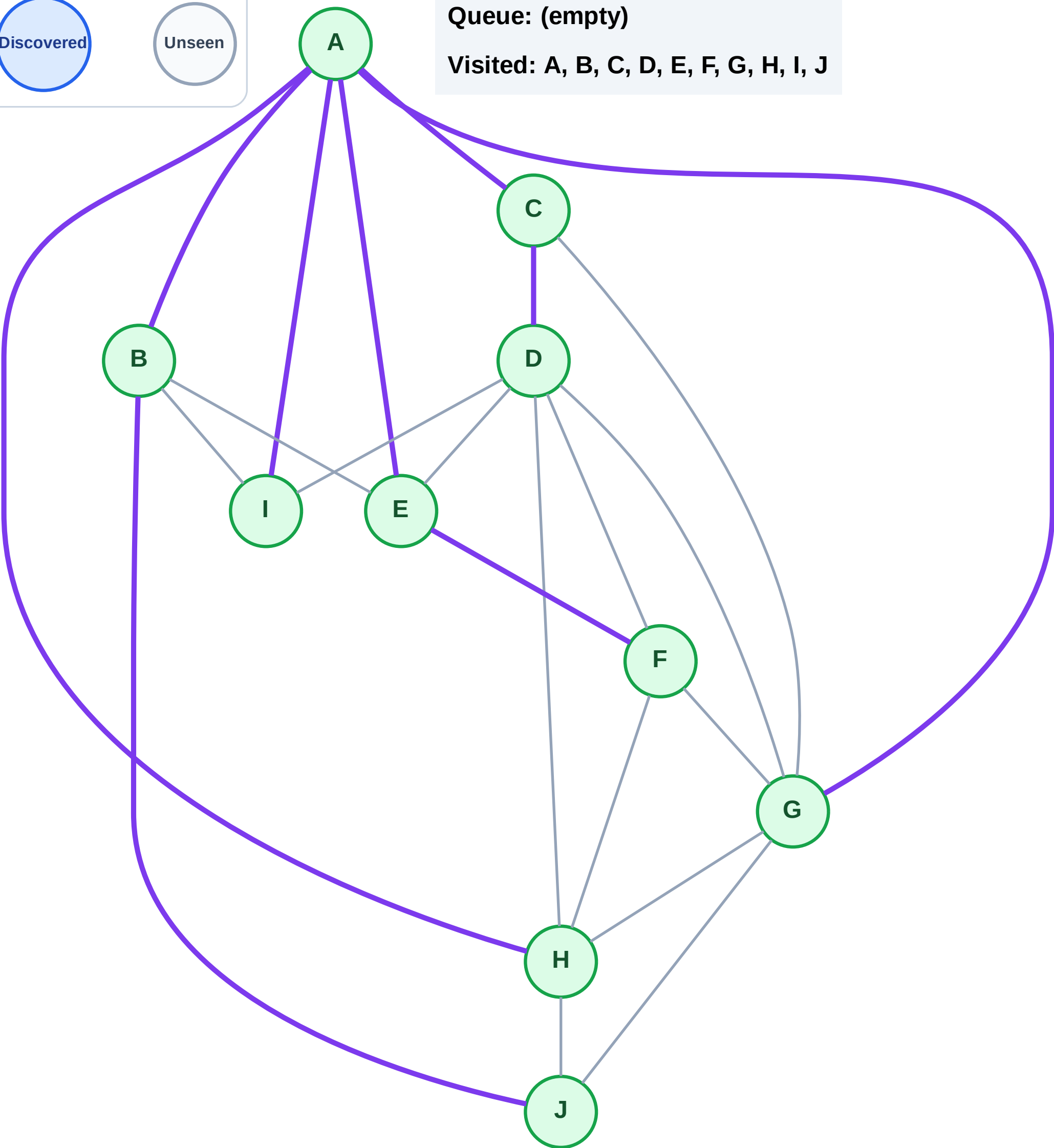
Step 21: No new neighbors from F

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

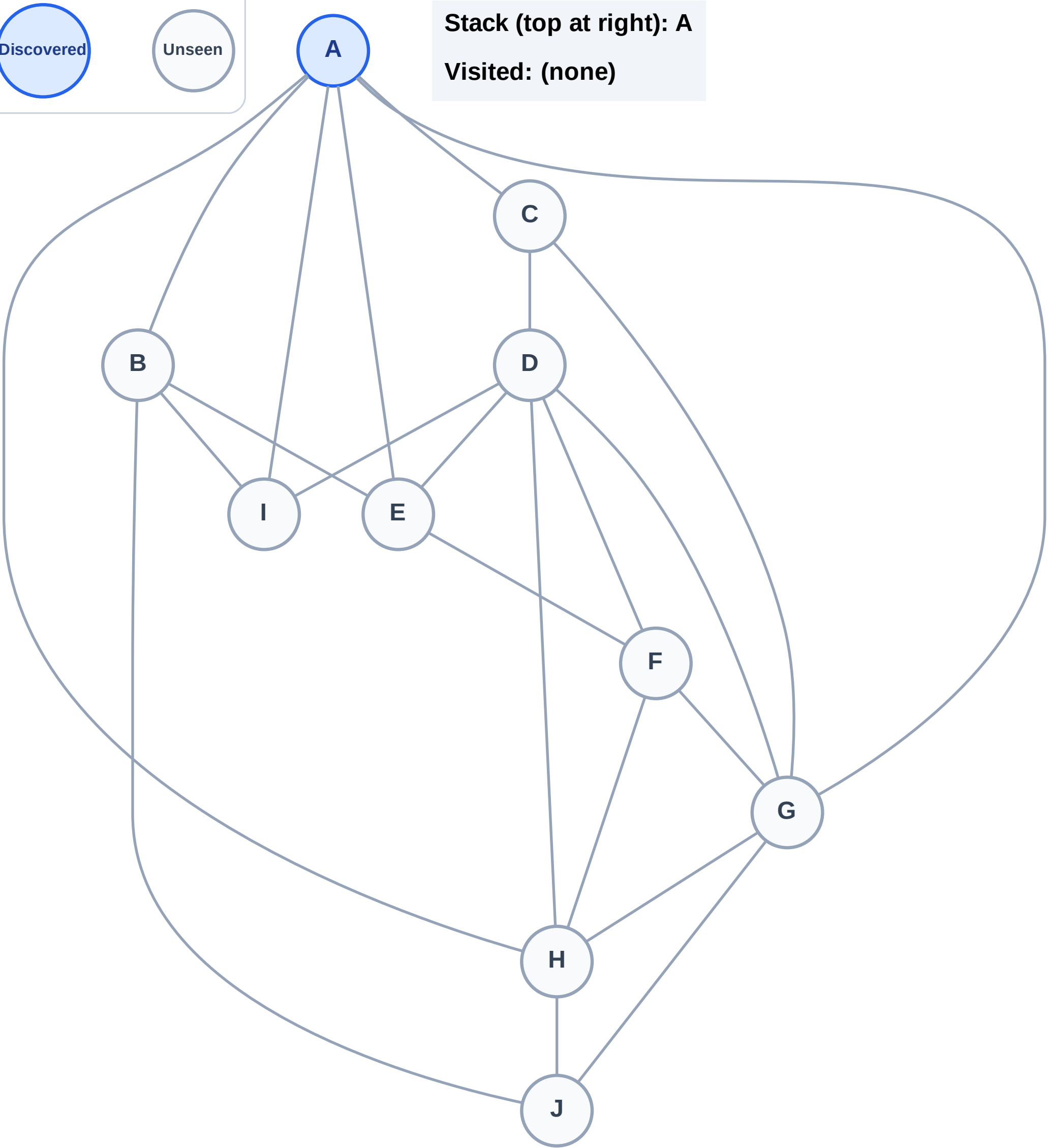
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

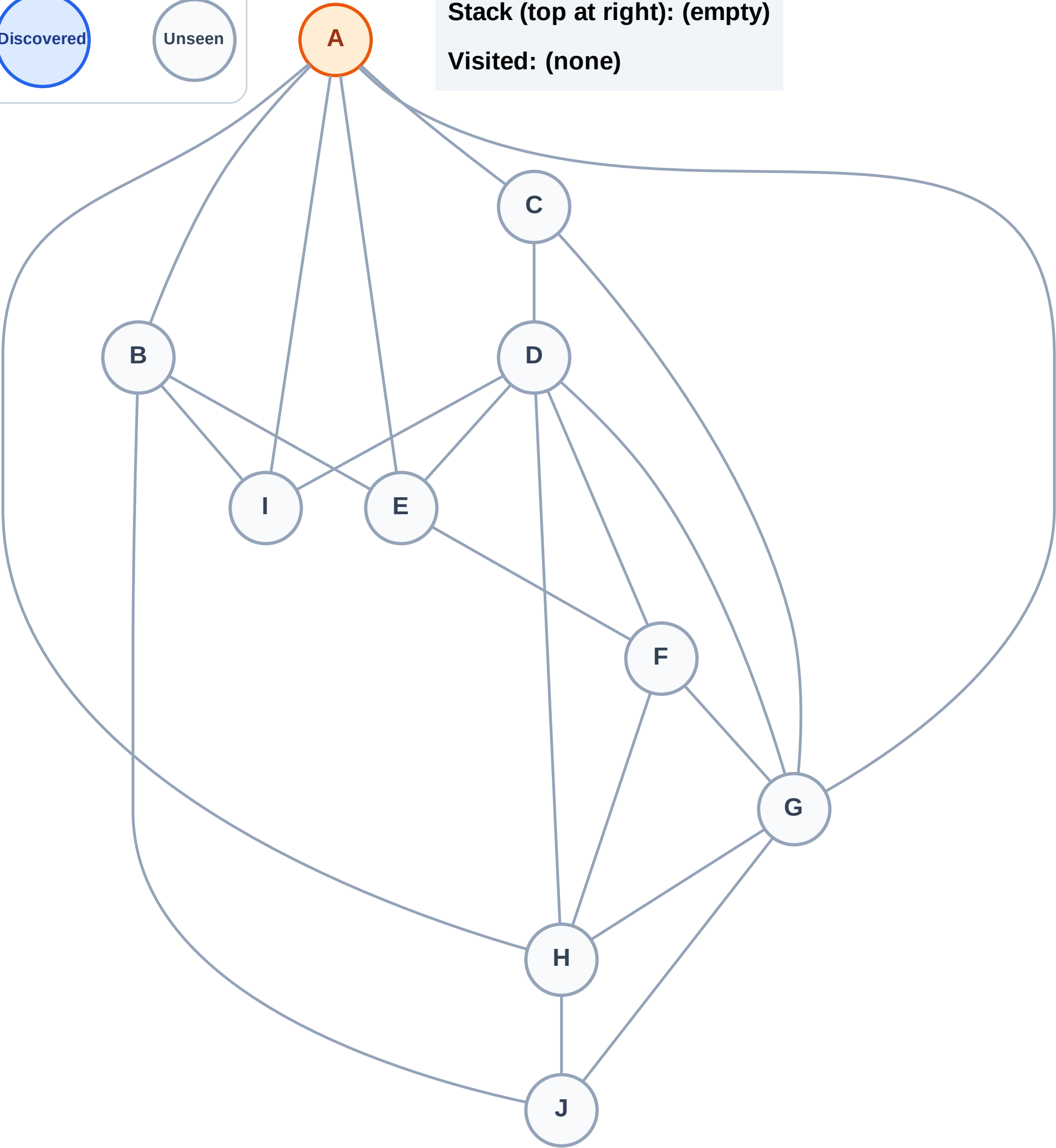
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

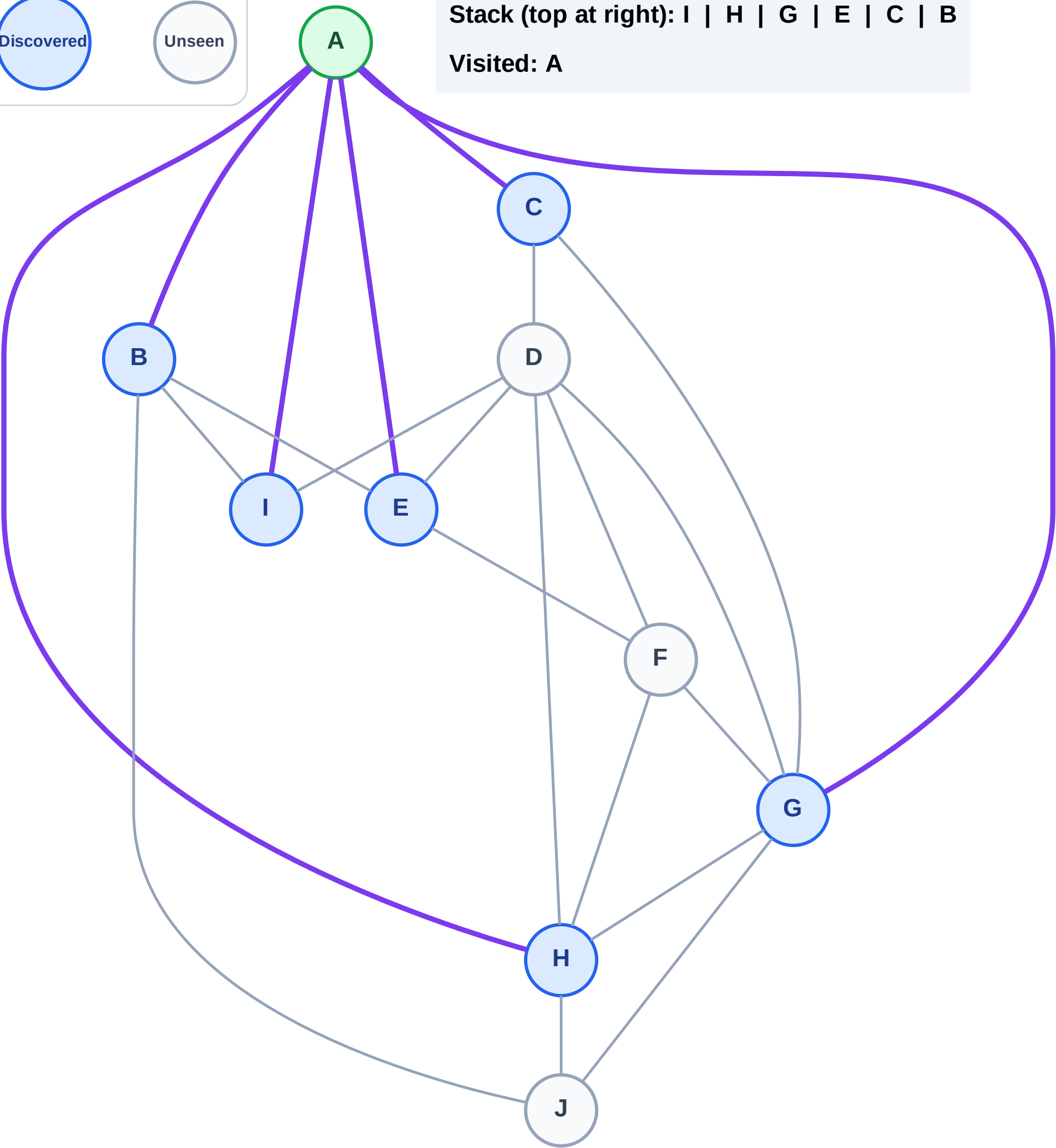
Step 3: Discover I, H, G, E, C, B from A

Legend



Stack (top at right): I | H | G | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

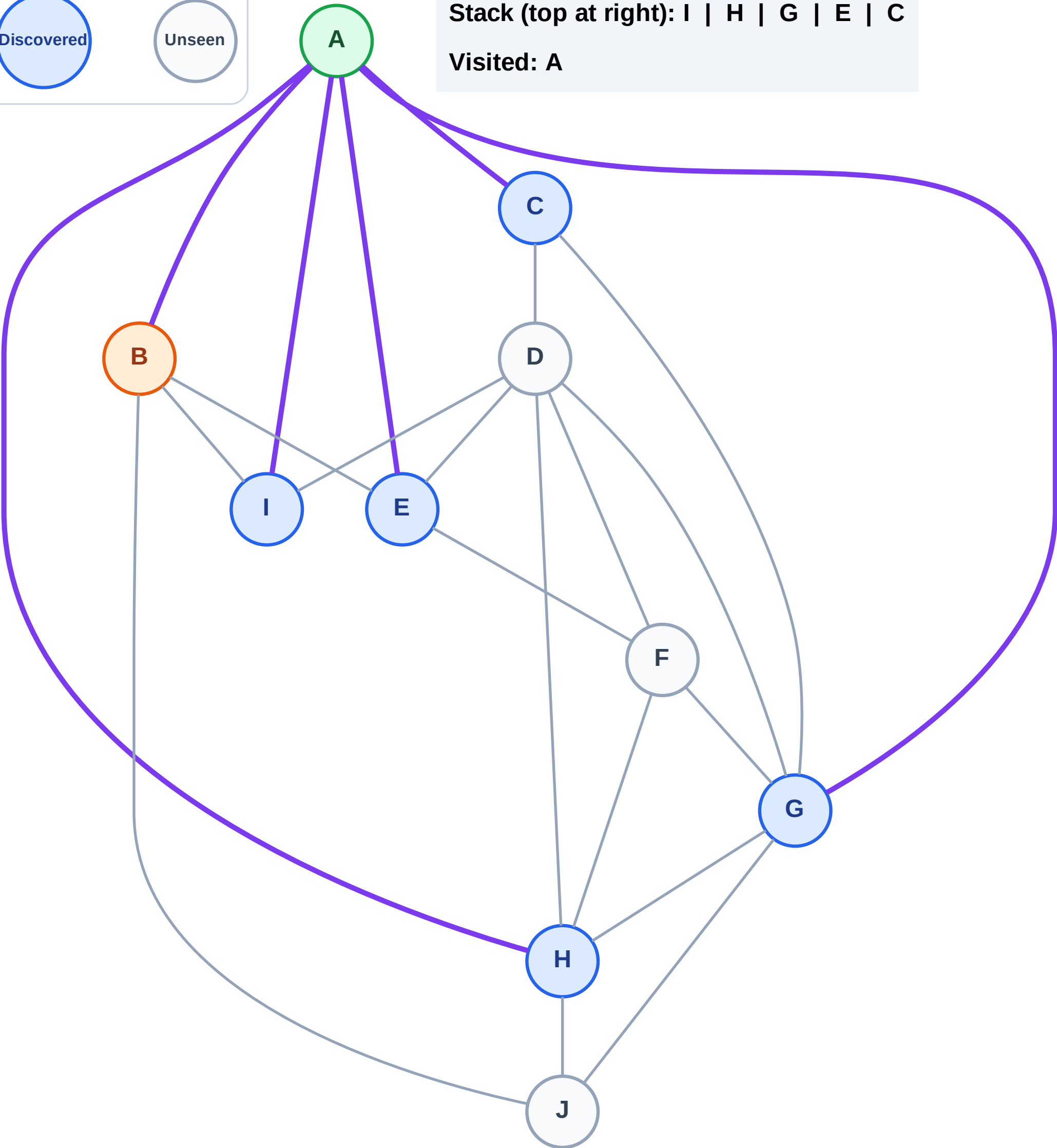
Processing

Discovered

Unseen

Stack (top at right): I | H | G | E | C

Visited: A



DFS Traversal

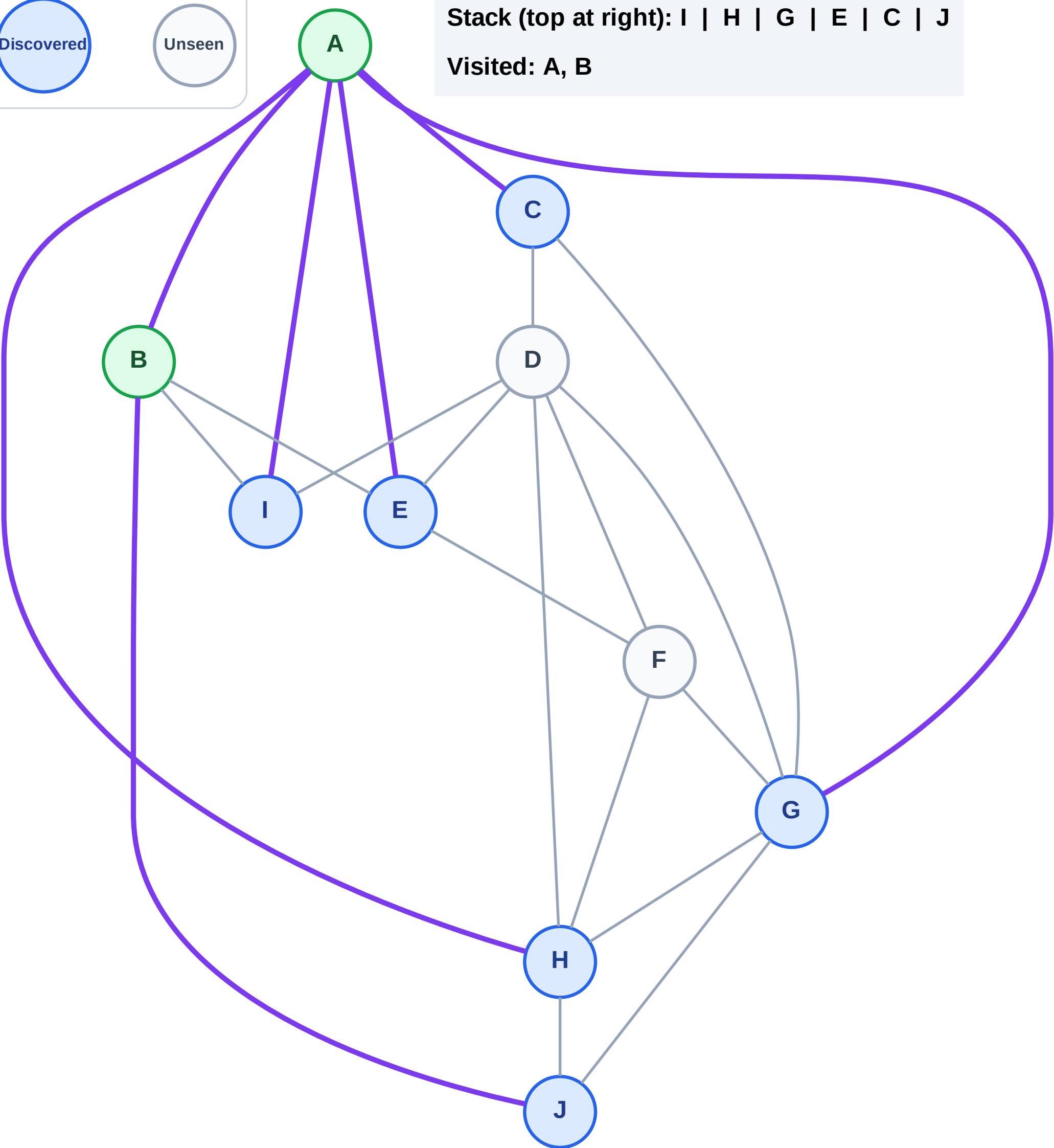
Step 5: Discover J from B

Legend



Stack (top at right): I | H | G | E | C | J

Visited: A, B



DFS Traversal

Step 6: Process node J

Legend

Complete

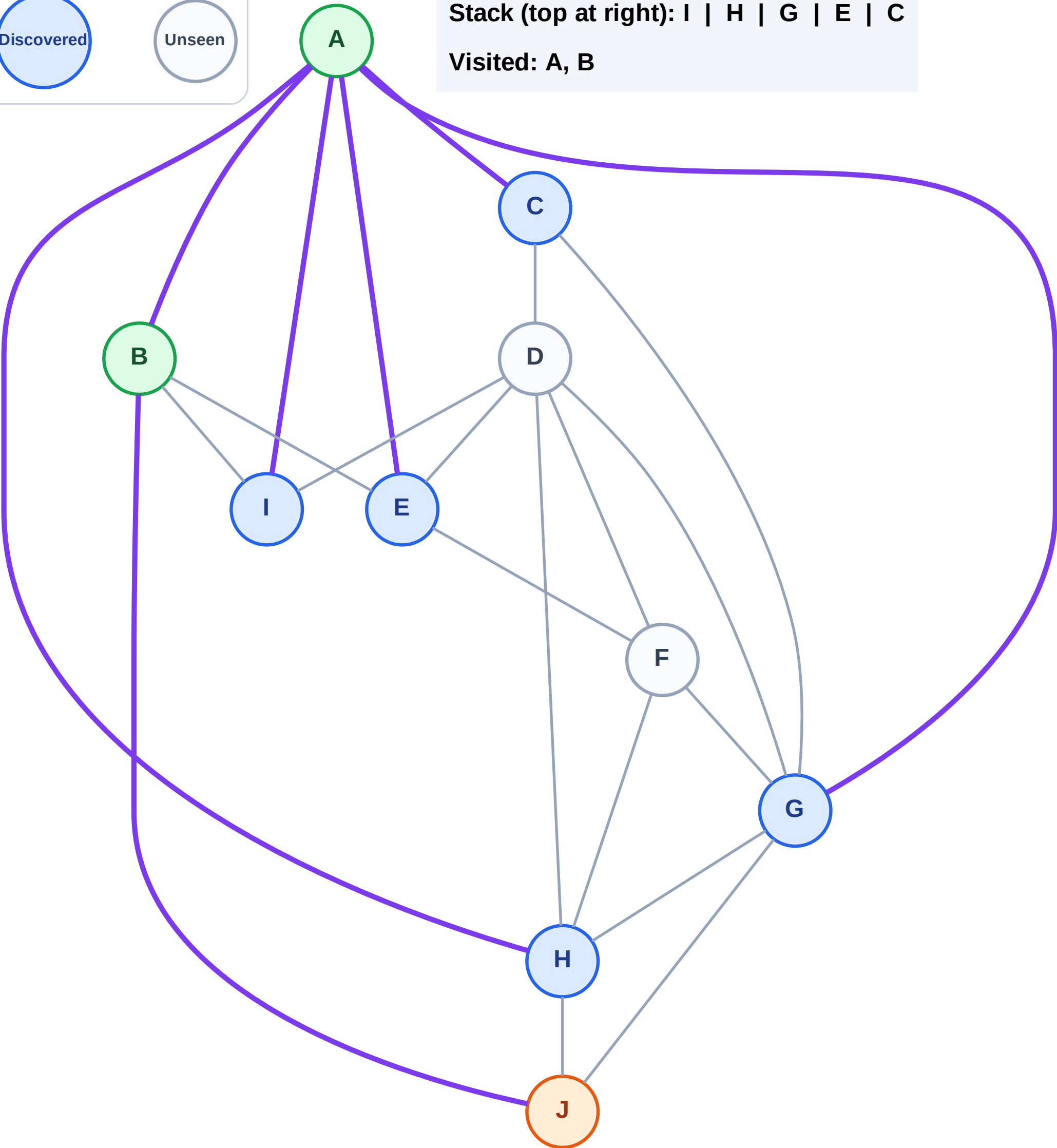
Processing

Discovered

Unseen

Stack (top at right): I | H | G | E | C

Visited: A, B



DFS Traversal

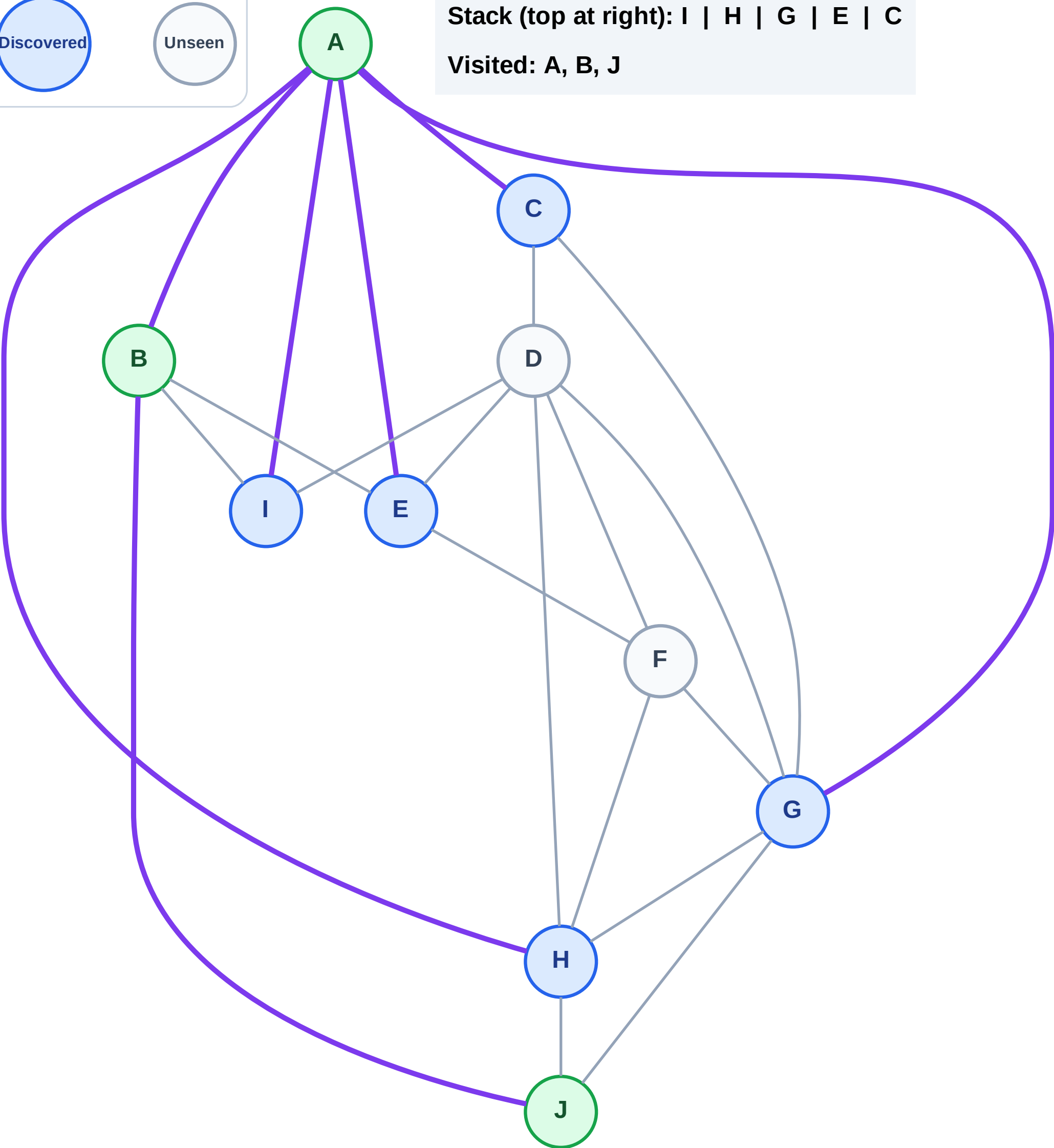
Step 7: No new neighbors from J

Legend



Stack (top at right): I | H | G | E | C

Visited: A, B, J



DFS Traversal

Step 8: Process node C

Legend

Complete

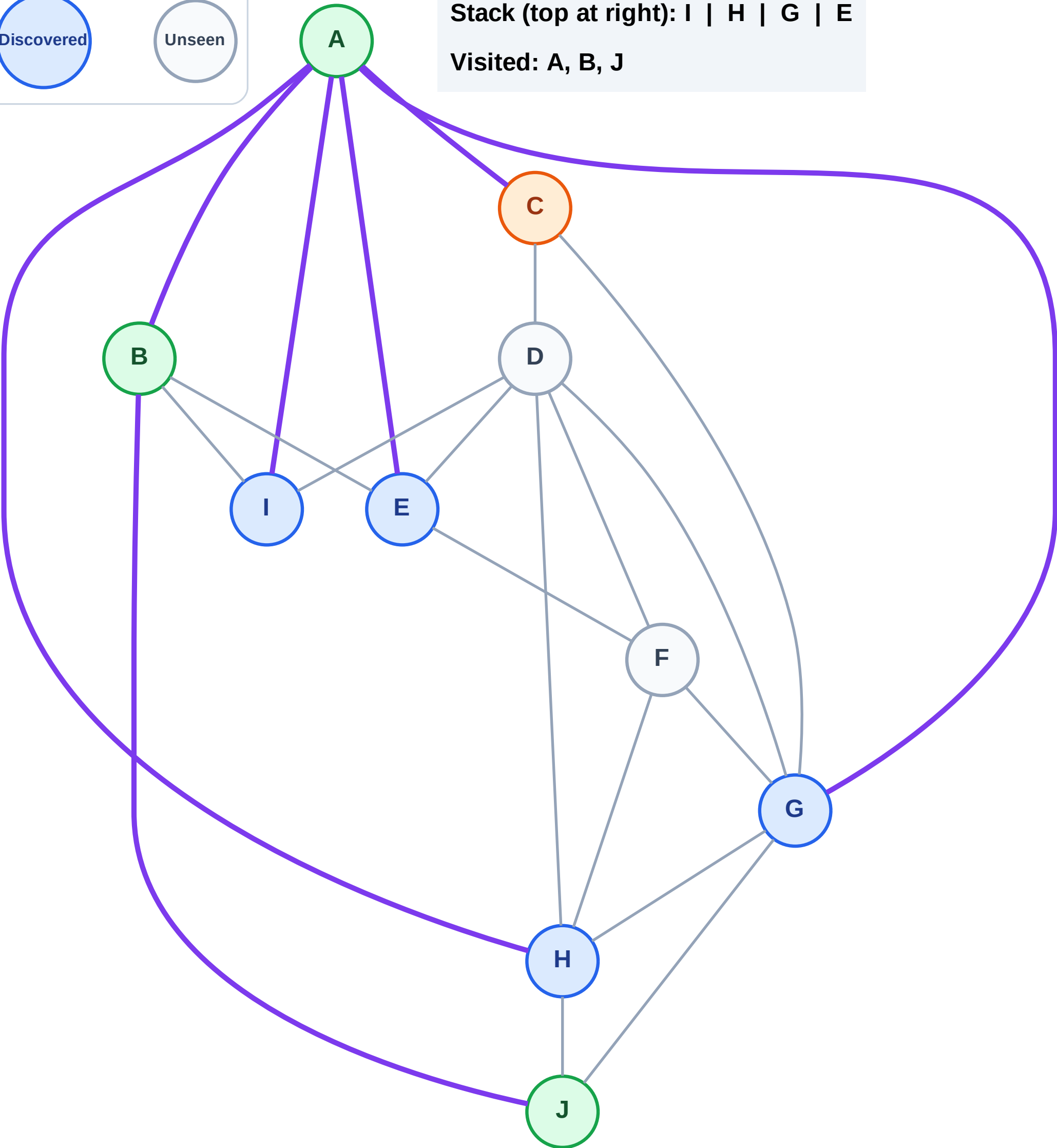
Processing

Discovered

Unseen

Stack (top at right): I | H | G | E

Visited: A, B, J



DFS Traversal

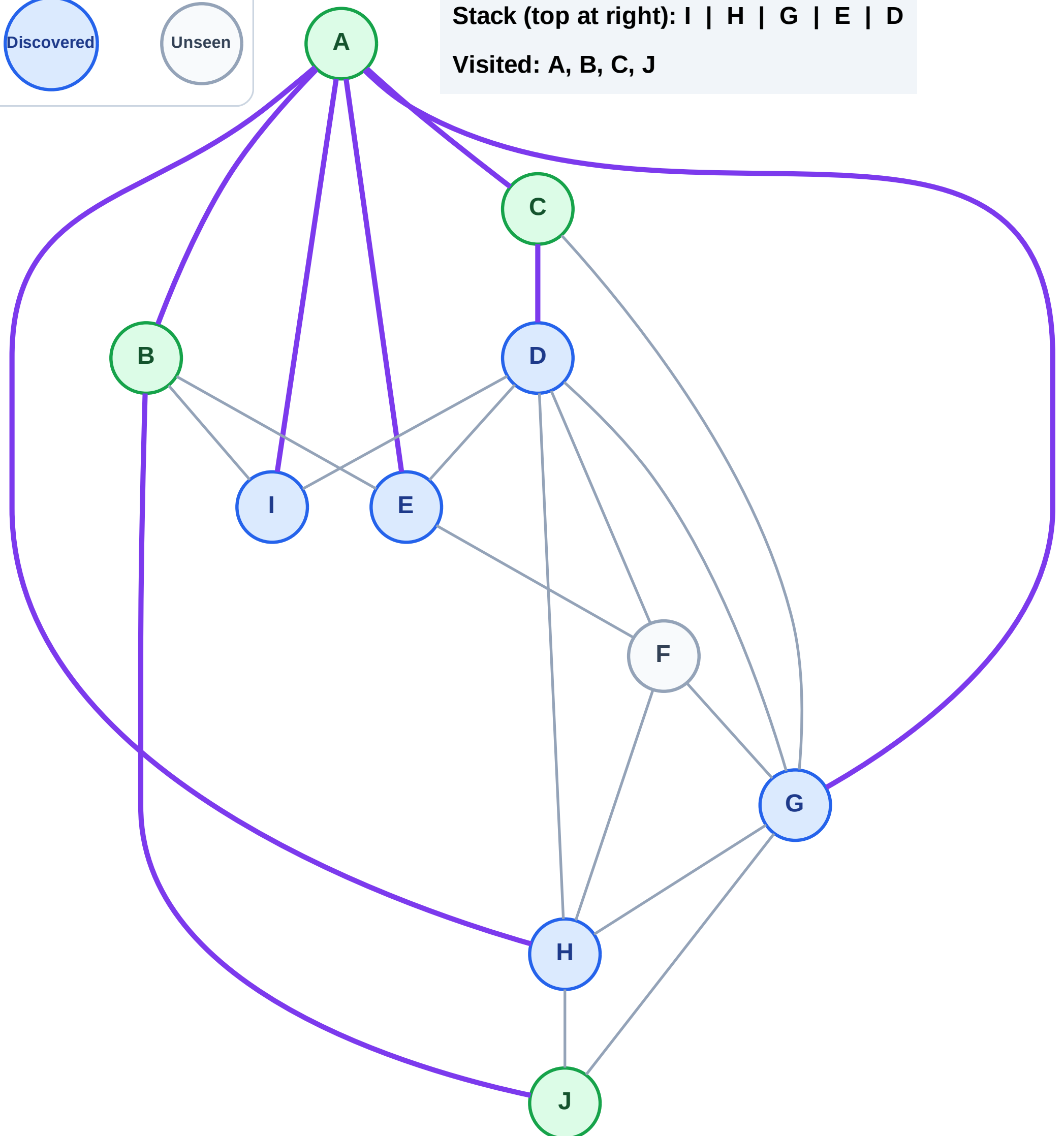
Step 9: Discover D from C

Legend



Stack (top at right): I | H | G | E | D

Visited: A, B, C, J



DFS Traversal

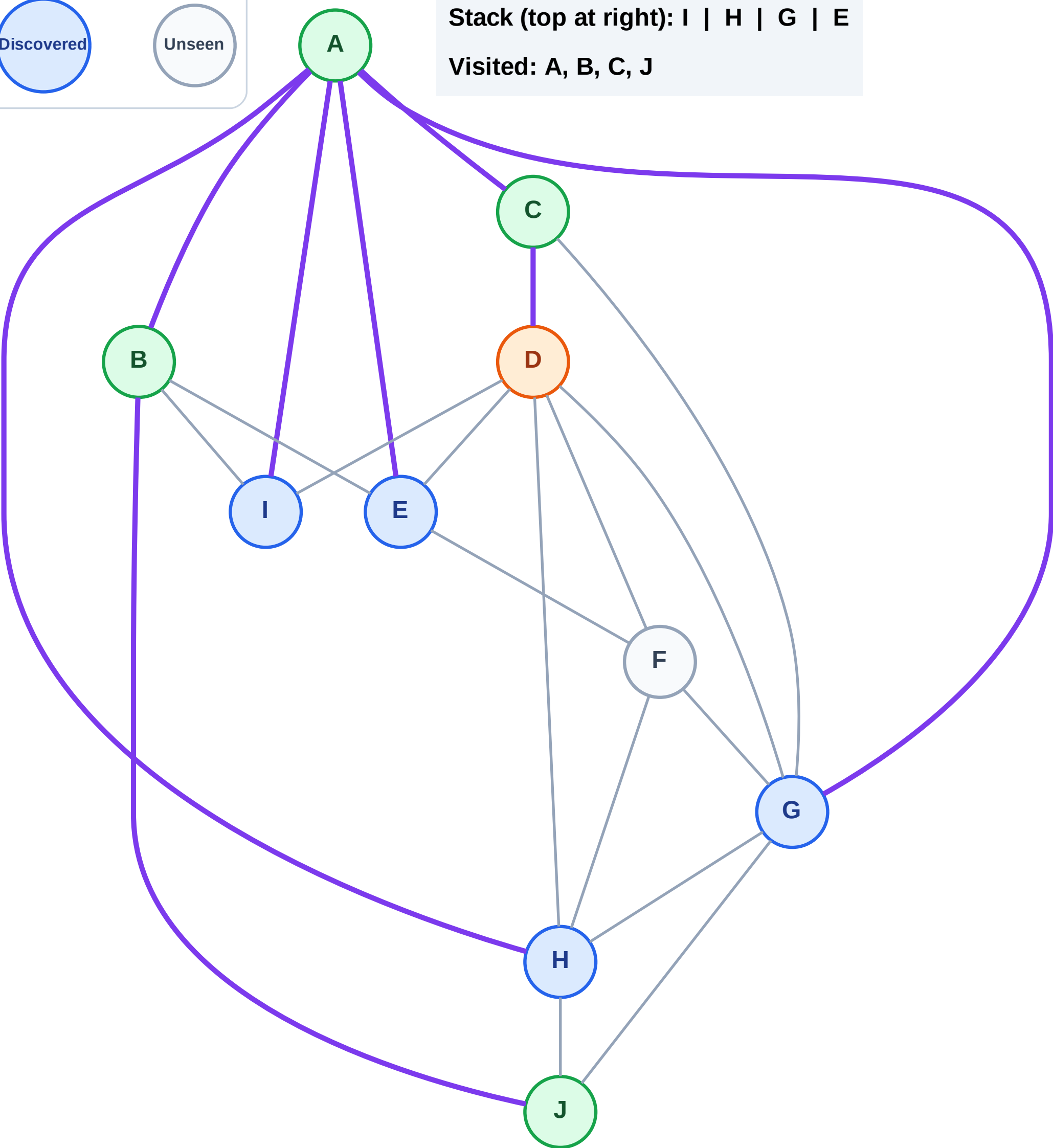
Step 10: Process node D

Legend



Stack (top at right): I | H | G | E

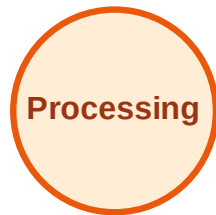
Visited: A, B, C, J



DFS Traversal

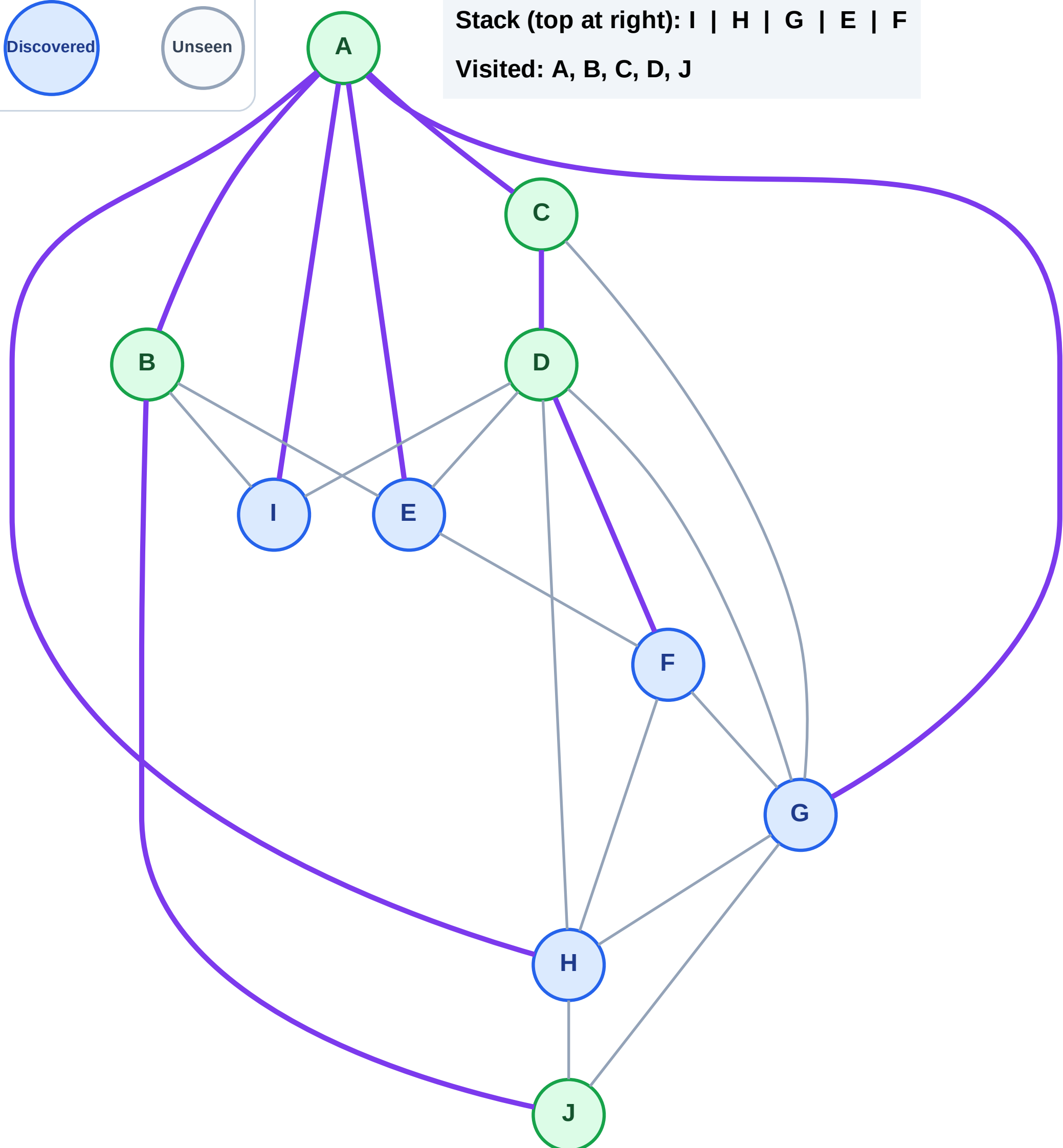
Step 11: Discover F from D

Legend



Stack (top at right): I | H | G | E | F

Visited: A, B, C, D, J



DFS Traversal

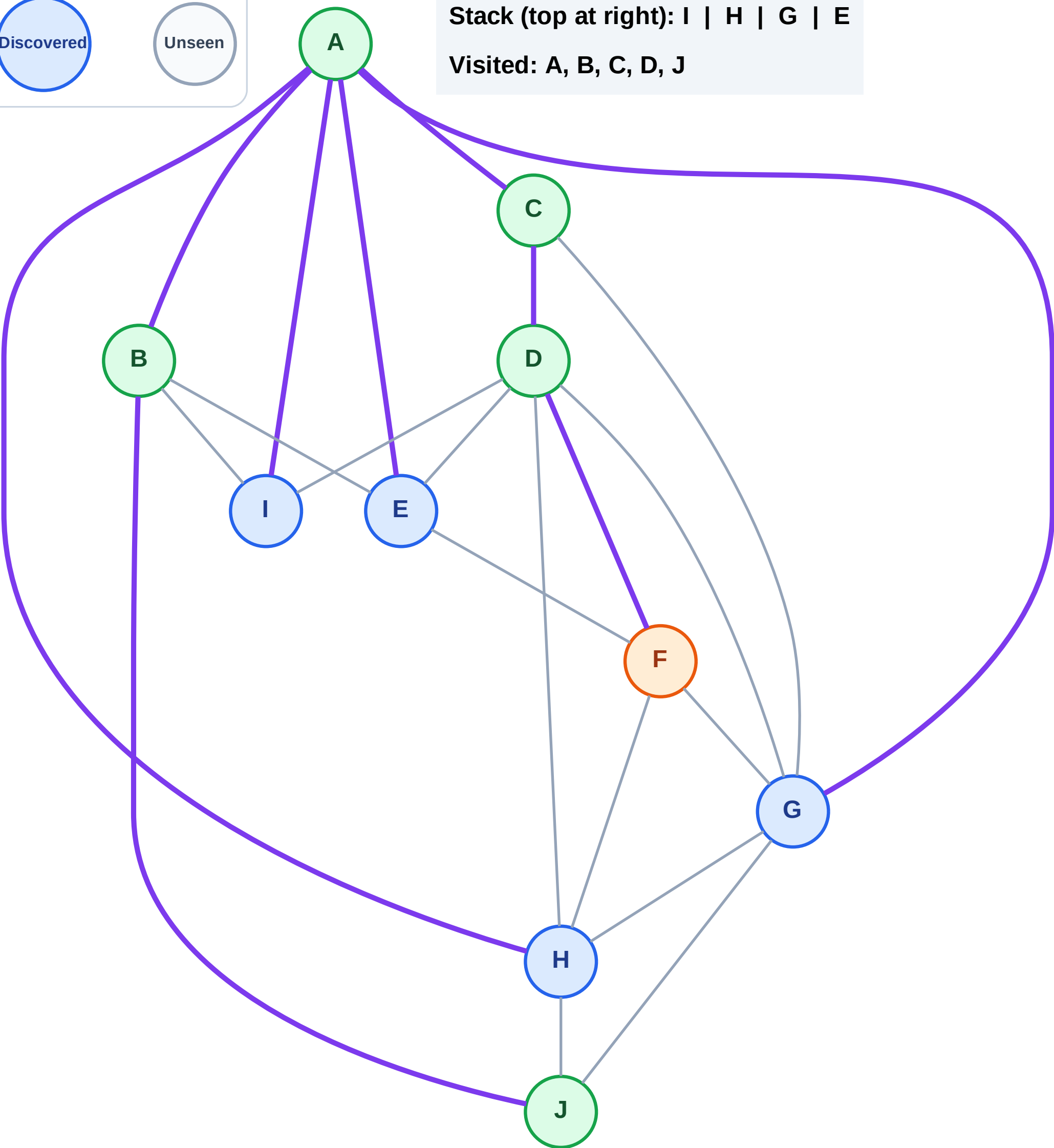
Step 12: Process node F

Legend



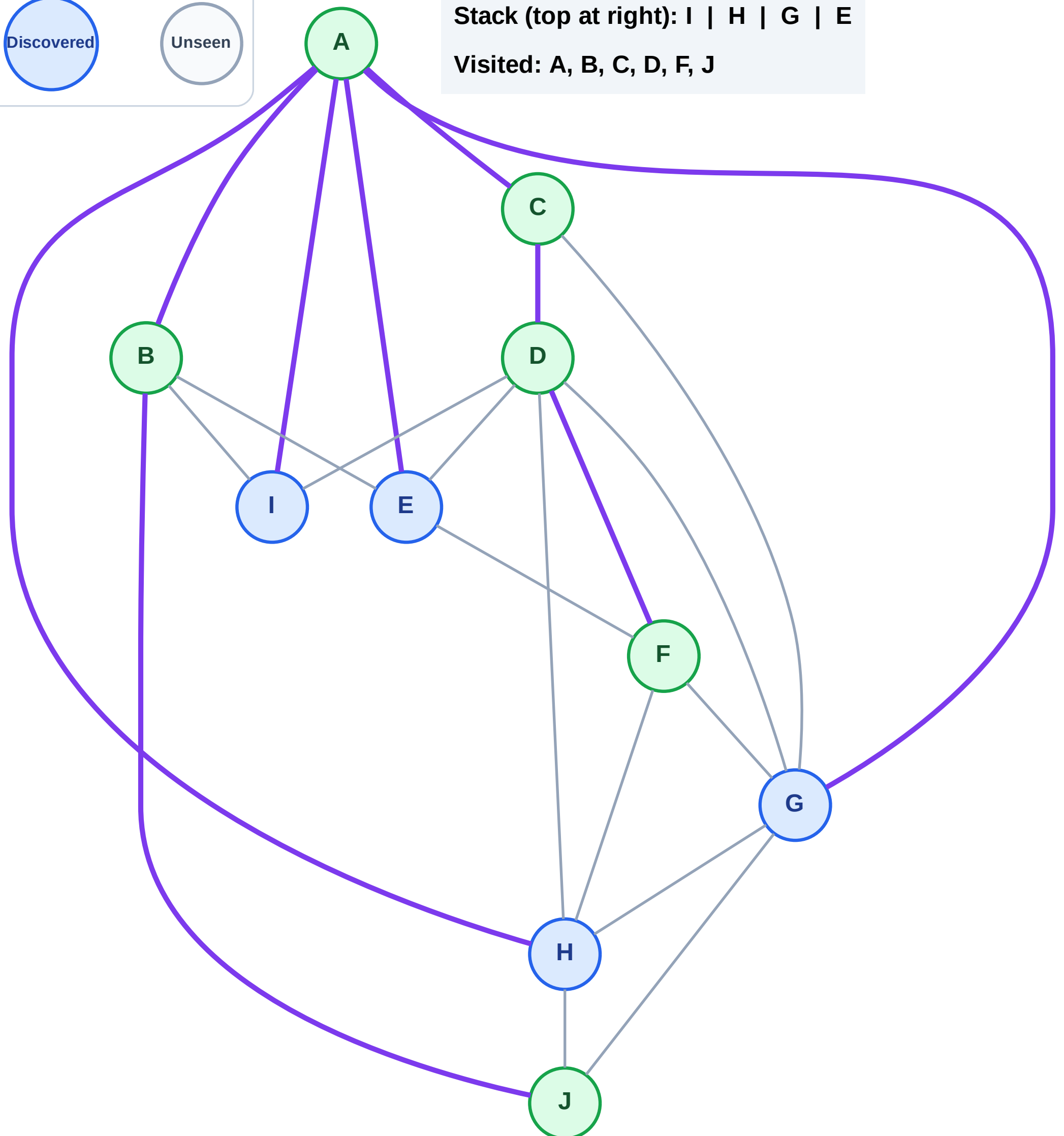
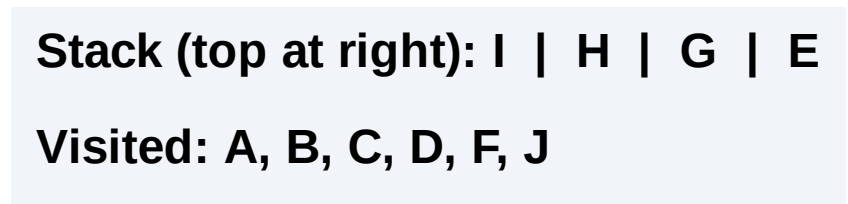
Stack (top at right): I | H | G | E

Visited: A, B, C, D, J



Step 13: No new neighbors from F

Step 13: No new neighbors from F



DFS Traversal

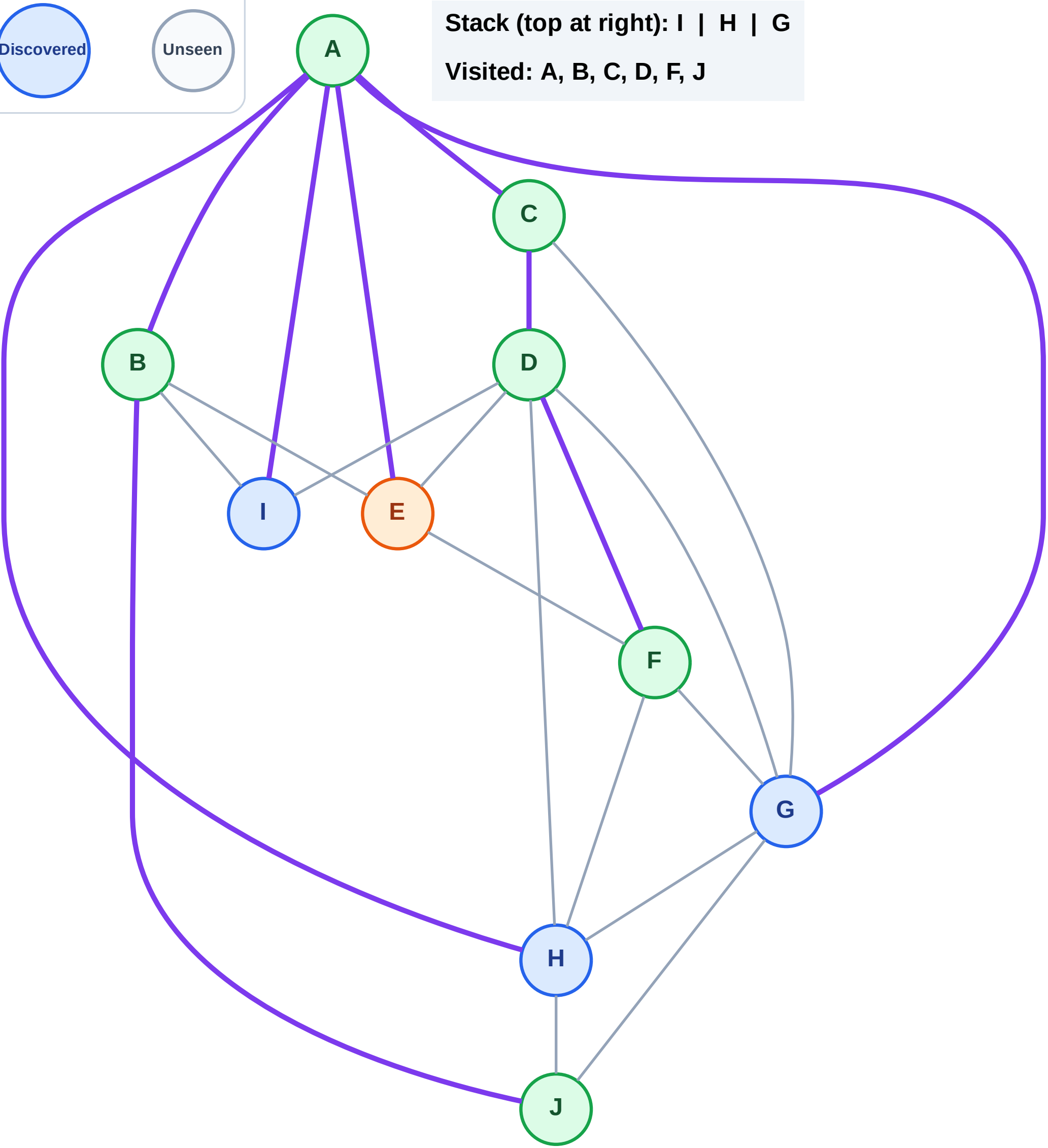
Step 14: Process node E

Legend



Stack (top at right): I | H | G

Visited: A, B, C, D, F, J



DFS Traversal

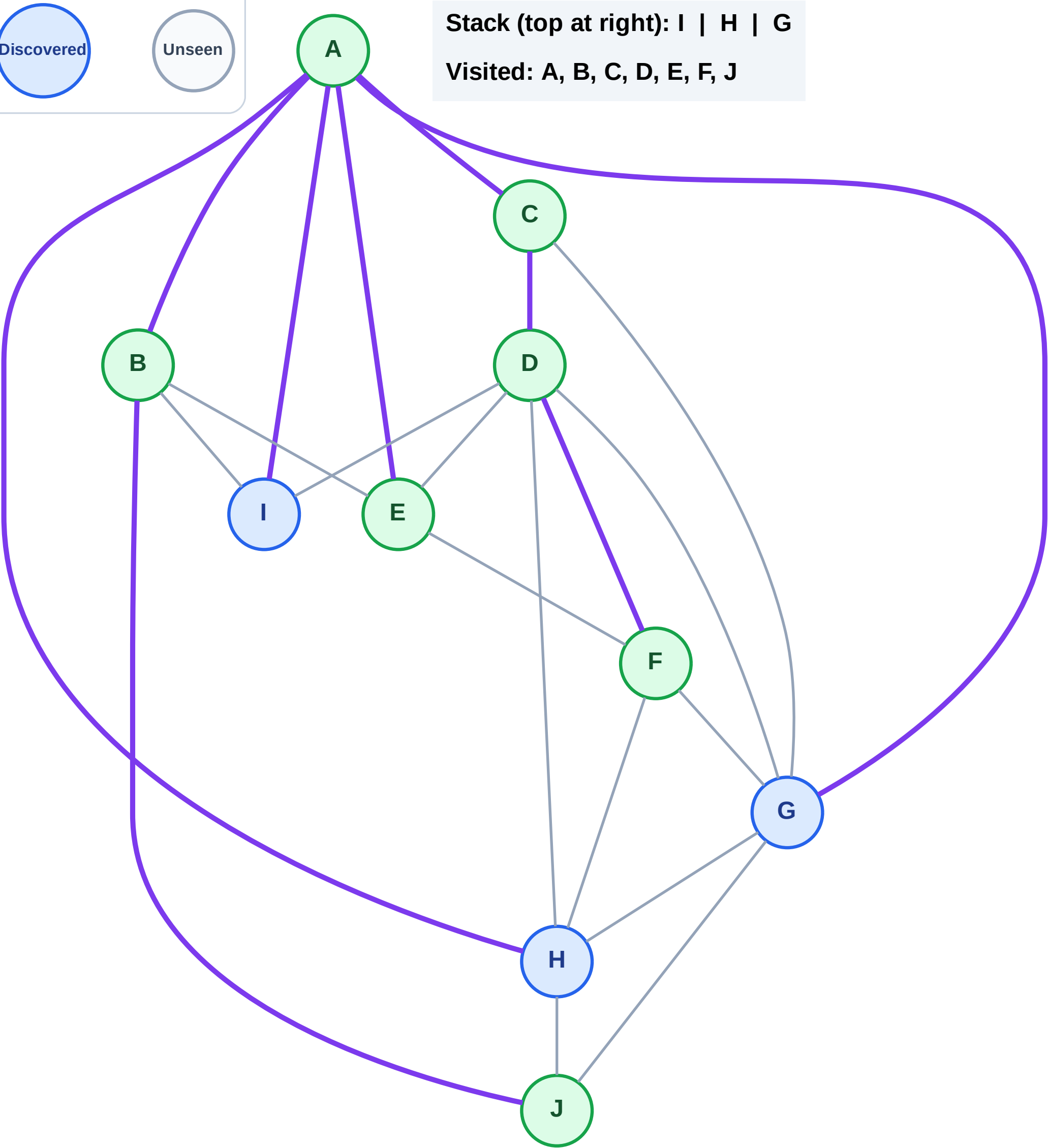
Step 15: No new neighbors from E

Legend



Stack (top at right): I | H | G

Visited: A, B, C, D, E, F, J



DFS Traversal

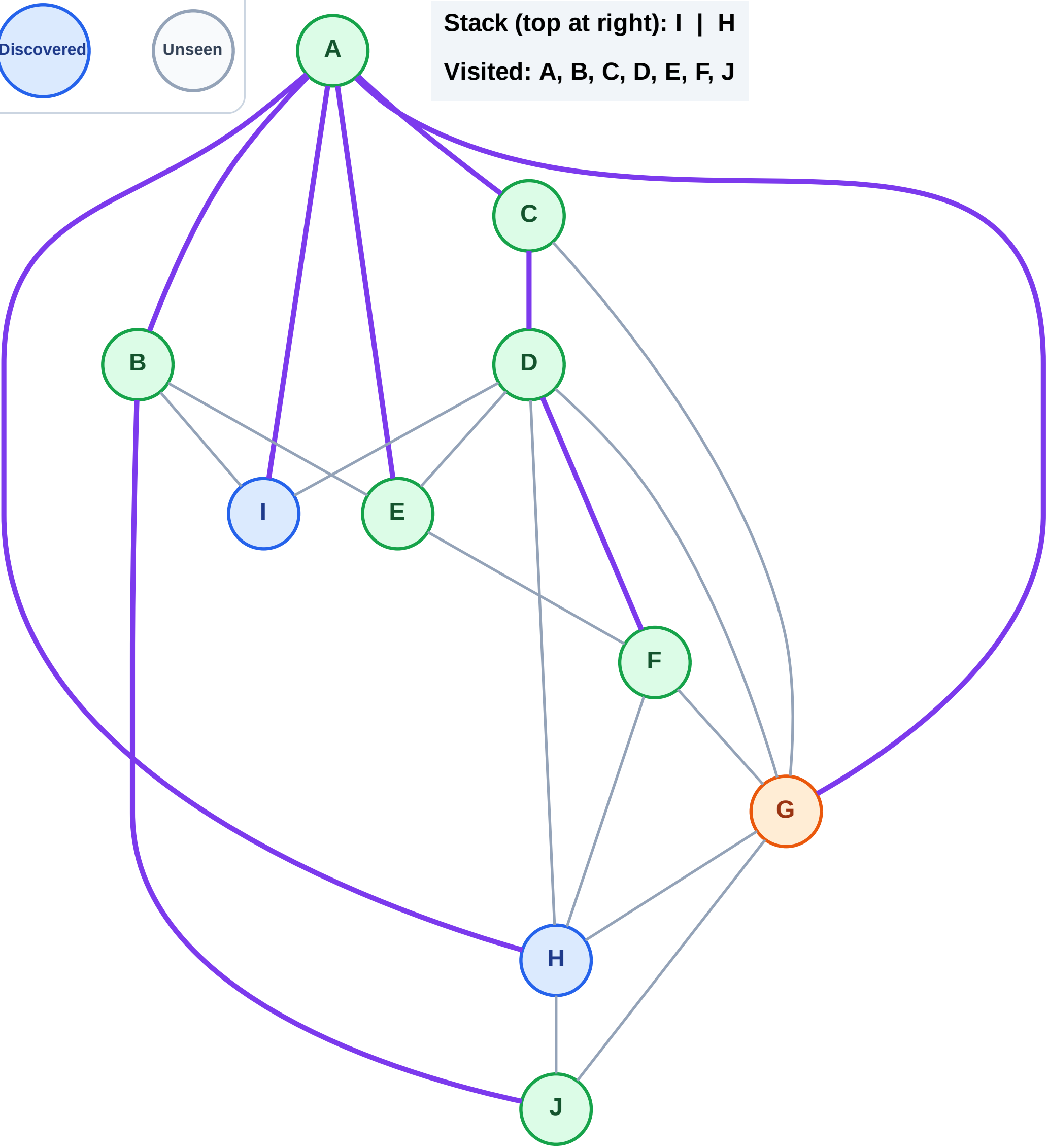
Step 16: Process node G

Legend



Stack (top at right): I | H

Visited: A, B, C, D, E, F, J



DFS Traversal

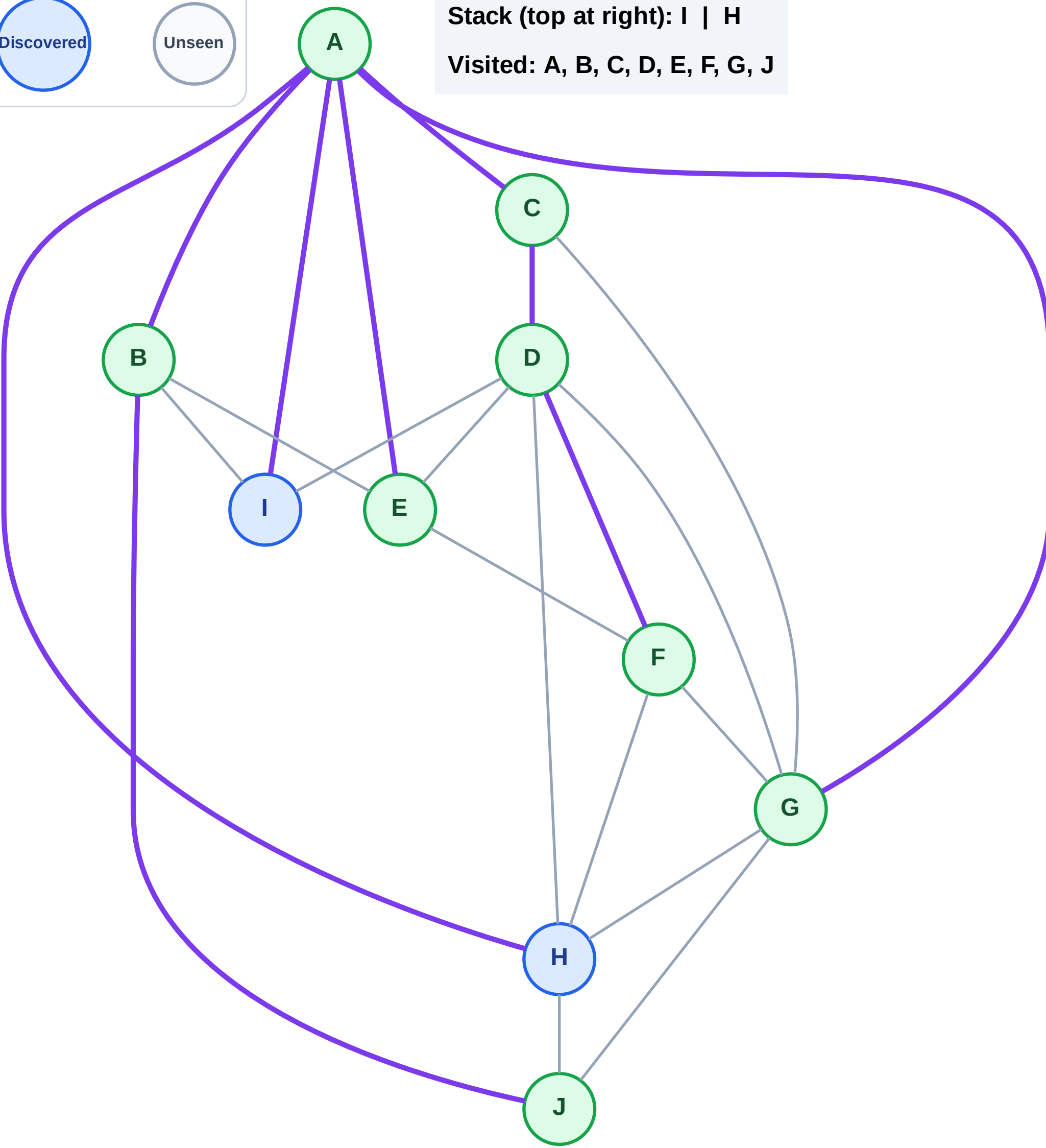
Step 17: No new neighbors from G

Legend



Stack (top at right): I | H

Visited: A, B, C, D, E, F, G, J



DFS Traversal

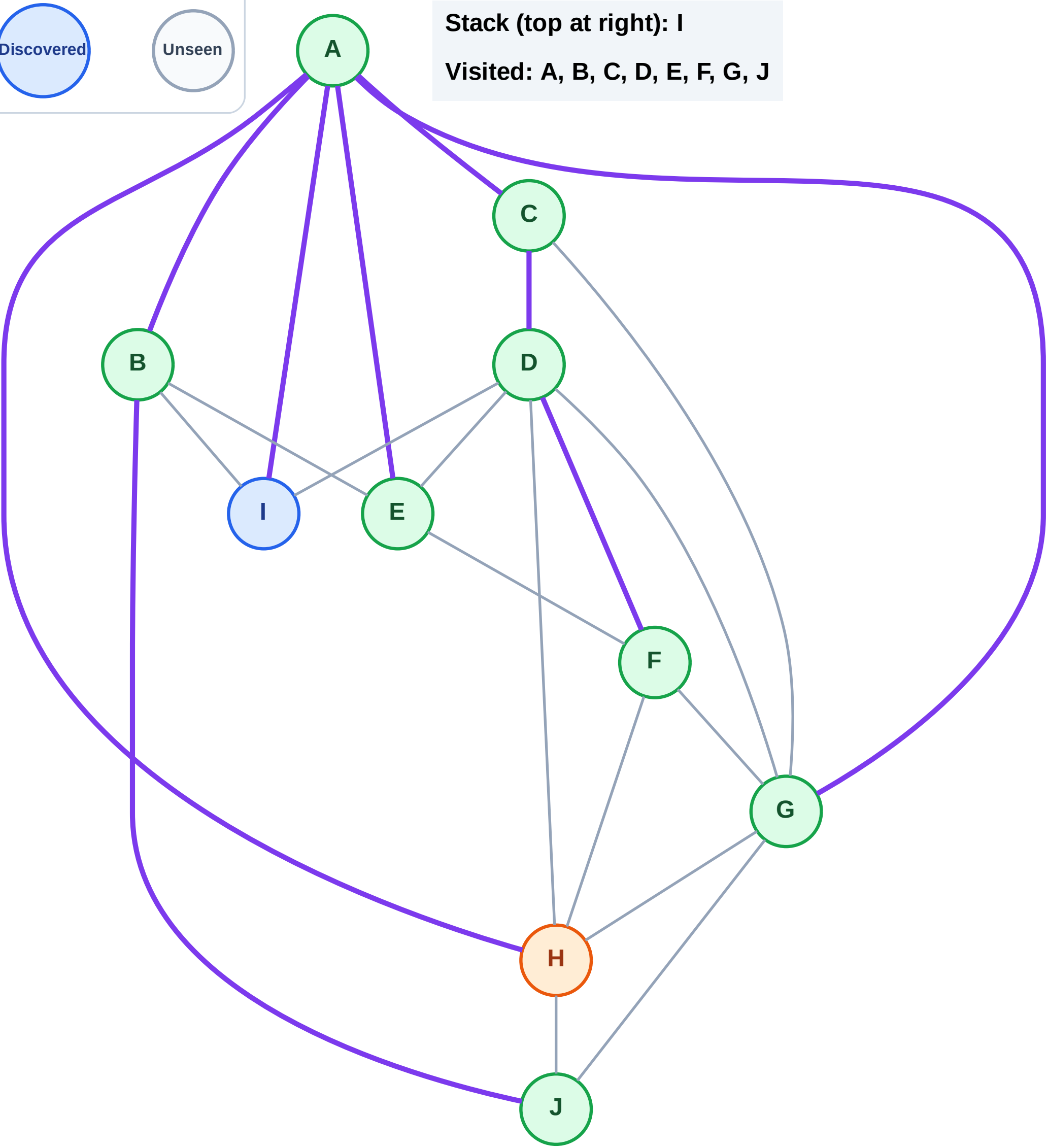
Step 18: Process node H

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, J



DFS Traversal

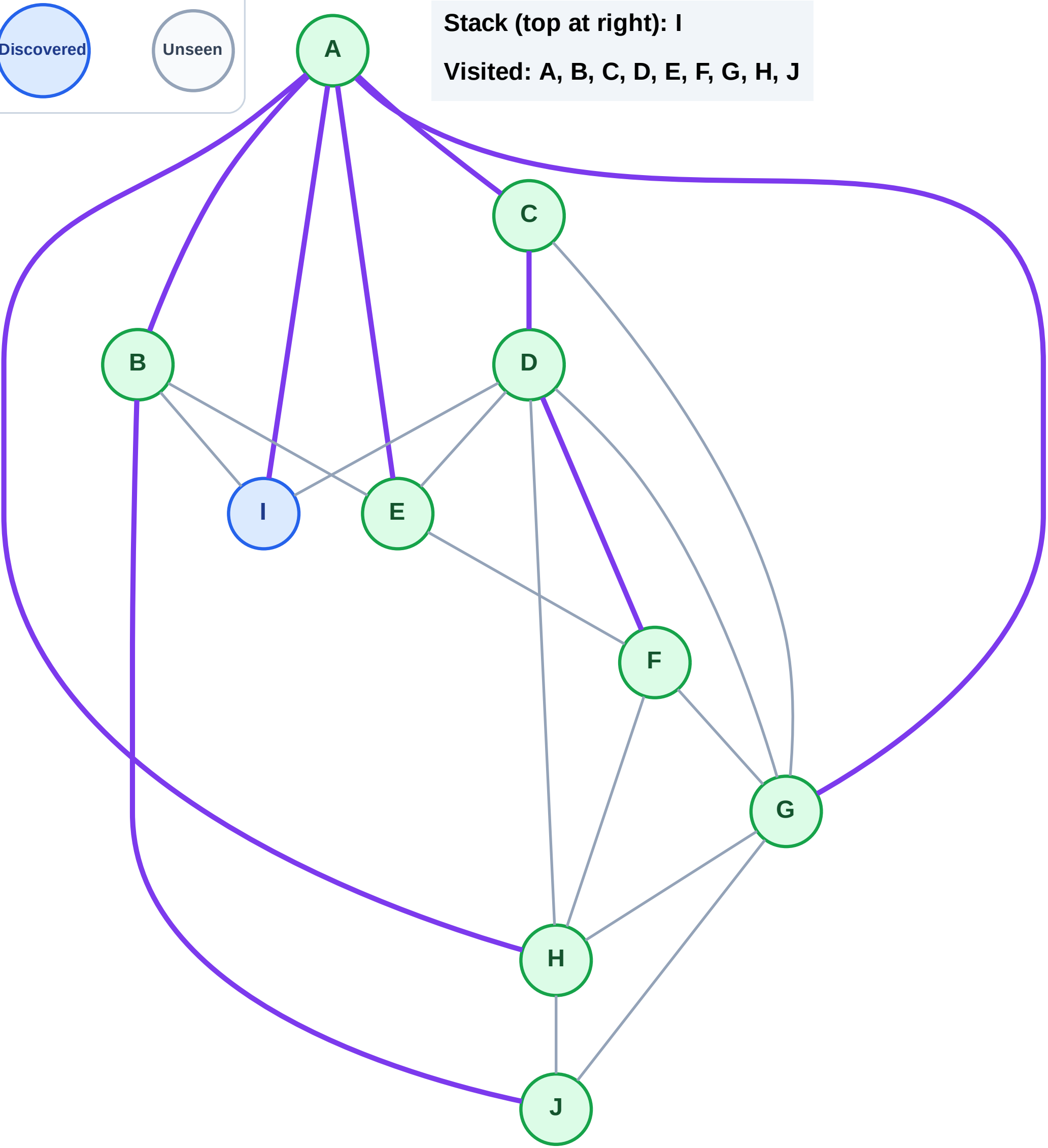
Step 19: No new neighbors from H

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 20: Process node I

Legend

Complete

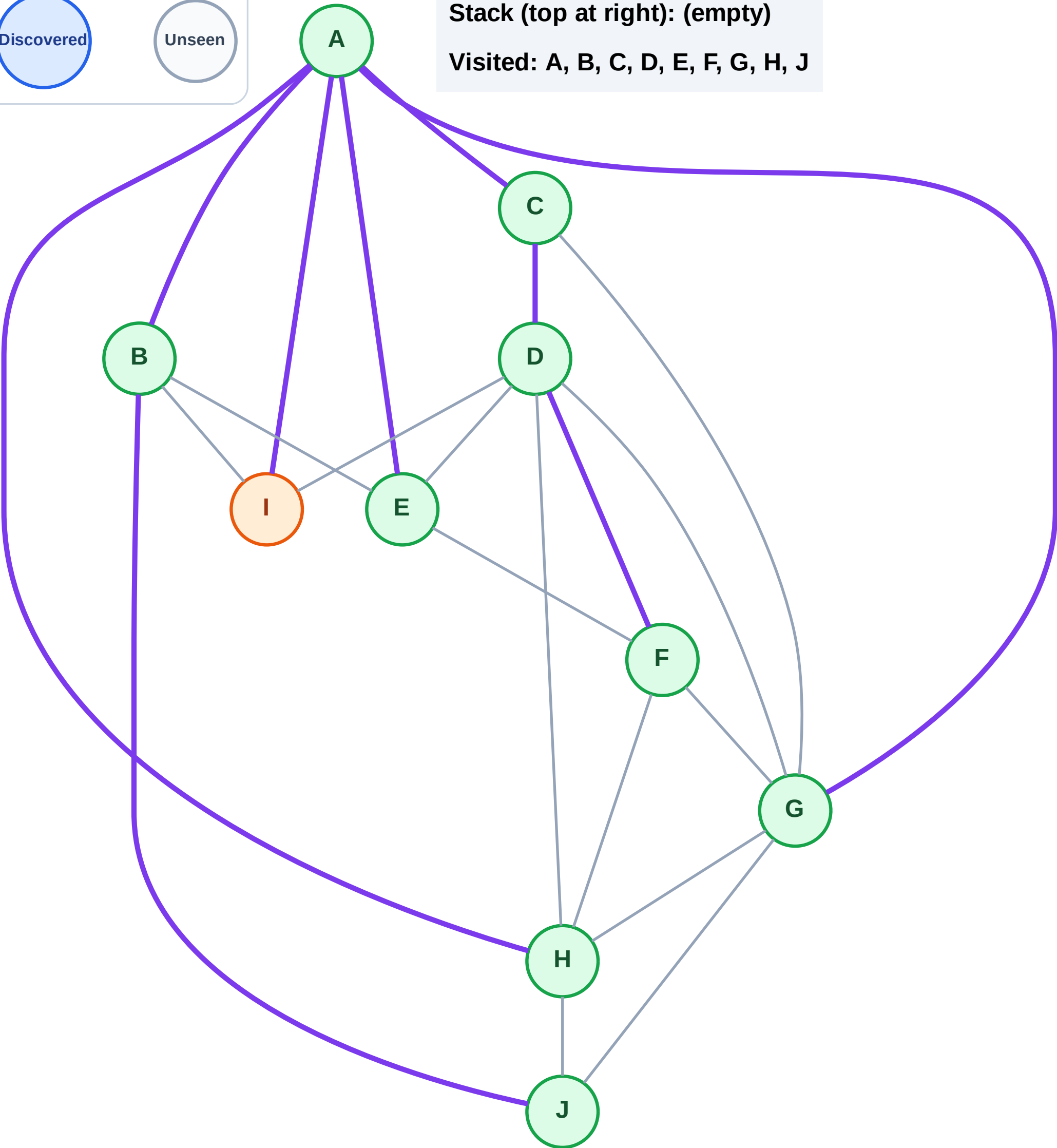
Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

